A Simple Approach to Staggered Difference-in-Differences in the Presence of Spillovers

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Abstract

We establish identifying assumptions and estimation procedures for the ATT in a Difference-in-Differences model with staggered treatment adoption in the presence of spillovers. We show that the ATT can be estimated by a simple TWFE regression that extends the approach of Wooldridge [2025]'s fully interacted regression. Moreover, we broaden our framework to nonlinear cases, offering estimation of the ATT by Poisson, Probit, and Logit regressions. We apply our method to revisit a corresponding application from the crime literature. Monte Carlo simulations suggest that our estimator performs competitively.

Keywords: Difference-in-Differences, staggered treatment adoption, spillovers, (non)linear models.

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1 Introduction

The Difference-in-Differences (DiD) literature, particularly the one concerned with staggered treatment adoption, has experienced significant advances in the last few years, and papers by Roth et al. [2023] and de Chaisemartin and D'Haultfœuille [2023] have summarized these developments. Within this array of advances, one area still understudied is the one linked to spillovers - implying that the Stable Unit Treatment Value Assumption (SUTVA) does not hold. However, as Roth et al. [2023] point out, spillover effects may be important in many economic applications, such as when a policy in one area affects neighboring areas, or when individuals are connected in a network. Our work contributes to this area and links two active DiD literature strands.

The first one focuses on estimation issues under staggered adoption and heterogeneous treatment effects across units and time, e.g., Borusyak et al. [2024], de Chaisemartin and D'Haultfoeuille [2020], Callaway and Sant'Anna [2021], Dube et al. [2025], Goodman-Bacon [2021], Sun and Abraham [2021] and Wooldridge [2025, 2023]. This literature highlights that the two-way fixed effect (TWFE) regression estimator may be biased for the average treatment effect on the treated (ATT), to the extreme of showing the opposite sign. The authors suggest alternative estimators that account for the variation in treatment timing, thereby providing a consistent estimator for the ATT. We contribute to this literature by extending it to the case of spillovers in both linear and nonlinear DiD settings.

The second DiD strand studies the identification of average treatment effects in the presence of spillovers, e.g., Berg et al. [2021], Butts [2023], Clarke [2017], Hettinger et al. [2023], Huber and Steinmayr [2021] and Xu [2023].[1] These studies highlight two main challenges for identification of the ATT if the treatment also impacts units that are not formally treated.

[1]Violations of the SUTVA assumptions are also discussed in the context of cross-section observational studies (e.g., Forastiere et al. [2021], Zigler et al. [2020]) and in the context of randomized controlled trials, (e.g., Aronow and Samii [2017], Han et al. [2024], Hudgens and Halloran [2008], Sävje et al. [2021], and Vazquez-Bare [2023]).

First, untreated units might no longer be valid controls. Proposed solutions mostly centre around ruling out spillovers for a given group of units, often based on some spatial or network distance, allowing the researcher to use this latter group as a control.[2] Alternatively, if sufficient information exists, one can parametrize how units are exposed to spillovers. Second, multiple definitions of the ATT exist in the presence of spillovers. This is because a unit's treatment can lead to changes to its own outcome, but also to other units' outcomes. In this case, the researcher might be interested in examining the former effect, summarized by the ATT without interference (i.e., the one identified under SUTVA), or in a broader definition of the ATT that also accounts for the latter effect. Here, we contribute to this literature by providing conditions that allow for the identification of the ATT without interference, despite the presence of spillovers. Moreover, we focus on the more complex staggered treatment adoption, offering consistent estimation of the ATT under staggered treatment adoption and the consequent evolution of spillover effects.

Specifically regarding contributions, first, we establish the identifying assumptions for the ATT without interference given a staggered DiD setting in the presence of spillovers. We show that, in addition to the canonical i) treatment irreversibility, ii) no-anticipation, and iii) parallel trends assumption, identification is established under the assumption that once a unit receives treatment, it is no longer influenced by spillover effects. This means that the unit forfeits any spillovers it may have previously received and remains unaffected by spillovers from subsequently treated groups. This assumption also makes the multiple definitions of the ATT equivalent, because they are the same with or without spillovers, simplifying policy evaluation and joining with the definition of ATT under SUTVA. In Section 3, we provide two examples in which such a scenario applies and discuss diagnostic tests to check the validity of the identifying assumptions. We further assume that a set of never-treated units is not exposed to spillovers, in line with the existing literature. The combination of

[2]This structure is sometimes referred to as partial interference.

these assumptions allows for the identification of the ATT. Moreover, we show that the average spillover effects on untreated units (ASUT) are also identified under a similar set of assumptions.

Our second contribution regards estimation. We extend the fully interacted TWFE regression approach of Wooldridge [2025] to account for spillovers.[3] Our approach applies to both linear and nonlinear parallel trends, offering estimation of the ATT and the ASUT by, for example, Probit, Logit and Poisson regressions, broadening the range of cases to which our approach can be applied.[4] For our empirical application, we revisit Gonzalez-Navarro [2013], who uses a nonlinear model to study the effects of installing a stolen vehicle recovery device on car theft, which is a count variable. Our proposed nonlinear estimator suggests a larger ATT relative to the original contribution's specification, which does not account for staggered treatment adoption. In addition, diagnostic tests suggest that our assumptions are likely to hold for the nonlinear model, but not for the linear model.

These contributions underscore the key distinctions between our work and Butts [2023], which appears to be most closely related. First, we provide assumptions for identifying the ATT without interference, whereas Butts [2023] identifies the sum of direct and spillover effects.[5] Furthermore, while Butts [2023] focuses on linear models, we extend our framework to both linear and nonlinear settings.

As a third and final contribution, we perform a Monte Carlo analysis, highlighting the bias-variance trade-off implicit in the correction for staggered treatment adoption and spillovers. Identification of time and group fixed effects can neither rely on the already treated units due to heterogeneous treatment effects, nor on the untreated units potentially

[3]This approach is numerically equivalent to the imputation approach of Borusyak et al. [2024].

[4]Roth and Sant'Anna [2023] point out that the parallel trends assumption is sensitive to the functional form, suggesting that the linear parallel trends assumption may not hold for outcomes such as binary and count data.

[5]When our key assumption does not hold, such that treated units are still exposed to spillover effects, our approach identifies the sum of direct and spillover effects, similarly to Butts [2023].

exposed to spillovers. However, the benefit of excluding such units from estimation can be small if treatment effects are relatively homogeneous and if spillovers are small, while costing the researcher precision. We compare the traditional TWFE estimator, which ignores both staggered adoption and spillovers, the Wooldridge [2025] estimator, which accounts for staggered adoption but not for spillovers, and our estimator, which corrects for both. We do so under different sample sizes, degrees of staggered treatment, and degrees of spillovers, showing that our estimator performs competitively in many settings.

The remainder of the paper is organized as follows. Section 2 lays out the formal DiD setup with staggered treatment adoption, after which Section 3 illustrates our setup with two motivating examples. Section 4 establishes conditions for identifying the ATT, while Section 5 discusses estimation and inference considering the formerly established assumptions. Section 6 extends our model to the nonlinear case, and Section 7 discusses a corresponding application. Section 8 concludes. The Appendix provides our Monte Carlo simulations.

2 Setup

We consider a DiD model with staggered treatment adoption, which involves panel data of units observed over time periods $t \in \{1, \ldots, T\}$. For each unit i at each time t, let D_{it} be the binary treatment status indicating whether the unit is treated (1) or not treated (0). We assume that the treatment is irreversible, meaning that once a unit undergoes treatment, it remains treated in all subsequent periods.

Assumption 1 (irreversibility)**.** *For any $s < t$, $D_{is} = 1$ implies $D_{it} = 1$.*

Under Assumption 1, we can categorize units into groups according to the periods at which they receive treatment for the first time. Let G_i be the group label of unit i, defined by $G_i \equiv \min\{t \mid D_{it} = 1\}$ if $D_{it} = 1$ for some t, and $G_i \equiv \infty$ if $D_{it} = 0$ for all t. We let $\mathcal{G}$ be the support of G_i. Under this group label, let $D_i \equiv (D_{i1}, \ldots, D_{iT})$ be the entire treatment

history of unit i. It follows that:

$$D_i = (\underbrace{0, \ldots, 0}_{t<g}, \underbrace{1, \ldots, 1}_{t \geq g}) \quad \text{if} \quad G_i = g, \tag{1}$$

where D_i equals a vector of zeros if $G_i = \infty$.

Let Y_{it} be the observed outcome of unit i at time t. Under SUTVA, it is standard to define the potential outcome by $Y_{it}(g)$, which is the outcome value when the own group label G_i is set to g. In the presence of spillover effects, this outcome is affected by the group labels of all units. We define the potential outcome by $Y_{it}(g_i, \mathbf{g}_{-i})$, where g_i is the group label of unit i and $\mathbf{g}_{-i}$ is the group labels of all units other than i in the population. In the case of finite population, $\mathbf{g}_{-i}$ will be a $(N-1)$ vector, and in the case of infinite population, $\mathbf{g}_{-i}$ will be a mapping from an index to $\mathcal{G}$. This notation does not impose any restrictions on the structure of the spillover effects. An alternative notation used in the literature (e.g., Vazquez-Bare, 2023 and Butts, 2023) is $Y_{it}(g_i, h_i(\mathbf{g}_{-i}))$, where $h_i(\cdot)$ is a deterministic mapping from $\mathbf{g}_{-i}$ to a non-negative real number (or vector) representing the spillover intensity. In this notation, $Y_{it}(g_i, \mathbf{g}_{-i})$ is obtained by setting h_i to be the identity mapping without loss of generality.

Similarly to the case of SUTVA, we define treatment effects by comparing the observed potential outcome against various counterfactual outcomes. Let G_i and $\mathbf{G}_{-i}$ be the observed group labels for unit i and the other units in the data, respectively, and define ∞ and ∞_{-i} as the group labels where unit i and the other units are all in group ∞, respectively.

The following four types of potential outcomes will be relevant to our discussion. First, $Y_{it}(G_i, \mathbf{G}_{-i})$ represents the outcome where unit i is treated according to its observed group label, and other units are treated according to their observed group labels. This potential outcome is observed, i.e., $Y_{it} = Y_{it}(G_i, \mathbf{G}_{-i})$. Second, $Y_{it}(G_i, \infty_{-i})$ represents the outcome where unit i is treated according to its observed group label, but all the other units are untreated. Third, $Y_{it}(\infty, \mathbf{G}_{-i})$ represents the outcome where unit i is untreated, but other

units are treated according to their observed group labels. Fourth, $Y_{it}(\infty, \infty_{-i})$ represents the outcome where both unit i and all the other units are untreated.

In the counterfactual outcomes defined above, we treat $\mathbf{G}_{-i}$ as a fixed design. More precisely, let $\mathbb{G}_{-i}$ denote a fixed object. We consider the four potential outcomes $Y_{it}(G_i, \mathbb{G}_{-i})$, $Y_{it}(G_i, \infty_{-i})$, $Y_{it}(\infty, \mathbb{G}_{-i})$, and $Y_{it}(\infty, \infty_{-i})$ and, if $\mathbf{G}_{-i} = \mathbf{g}_{-i}$ in the observed data, we set $\mathbb{G}_{-i} = \mathbf{g}_{-i}$. This interpretation is particularly intuitive in the simple case where potential outcomes take the form $Y_{it}(G_i, h(\mathbf{G}_{-i}))$, where $h \in [0, 1]$ represents spillover intensity. In this case, defining $h_i \equiv h(\mathbf{G}_{-i})$ as unit i's observed spillover intensity, the four potential outcomes evaluate spillover intensity either at its observed level h_i or at zero. For brevity of notation, we abuse notation and write $\mathbf{G}_{-i}$ in place of $\mathbb{G}_{-i}$. Accordingly, for the remainder of the paper, we write expressions such as $\mathbb{E}(Y_{it}(g, \mathbb{G}_{-i})|G_i = g)$ as $\mathbb{E}(Y_{it}(g, \mathbf{G}_{-i})|G_i = g)$, where $\mathbf{G}_{-i}$ denotes unit i's fixed observed design.

We assume no anticipatory effect exists for the four types of potential outcomes, a standard assumption in DiD analyses. Let $q \equiv \min\{t \mid t \in \mathcal{G}\}$ be the first period that any unit enters treatment.

Assumption 2 (no anticipation)**.** *For every i and for each $\mathbf{g}_{-i} \in \{\mathbf{G}_{-i}, \infty_{-i}\}$:*

$$Y_{it}(G_i, \mathbf{g}_{-i}) = Y_{it}(\infty, \infty_{-i}) \quad \textit{for every } t < q, \textit{ and}$$
$$Y_{it}(G_i, \mathbf{g}_{-i}) = Y_{it}(\infty, \mathbf{g}_{-i}) \quad \textit{for every } q \leq t < G_i.$$

Next, we introduce the parallel trends assumption for a linear DiD model, following the assumption in the absence of spillovers in Borusyak et al. [2024] and Wooldridge [2023]. Later, we will extend it to nonlinear parallel trends.

Assumption 3 (parallel trends, linear model)**.** *For every group g and time t,*

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = g) = \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty).$$

Remark 1. An equivalent characterization of Assumption 3 is

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = g) = \mathbb{E}(\alpha_i|G_i = g) + \delta_t, \tag{2}$$

where $\alpha_i \equiv Y_{i1}(\infty, \infty_{-i})$ and $\delta_t \equiv \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty)$. This characterization is more commonly written as

$$Y_{it}(\infty, \infty_{-i}) = \alpha_i + \delta_t + \varepsilon_{it}, \tag{3}$$

where

$$\varepsilon_{it} \equiv Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i}) - \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty),$$

which satisfies $\mathbb{E}(\varepsilon_{it}|G_i = g) = 0$ for all $g \in \mathcal{G}$ under Assumption 3. We refer to δ_t as the common time effect.

In the presence of spillover effects, multiple definitions of the ATT arise (Huber and Steinmayr [2021], Hudgens and Halloran [2008], Sävje et al. [2021]). We first introduce the ATT without interference (Huber and Steinmayr [2021], Vazquez-Bare [2023]):

$$ATT_0(g, t) \equiv \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g).$$

This definition of ATT captures the expected treatment effect at time t when unit i is the only treated unit in the population, thereby excluding any spillover effects from the other units. Alternatively, using the notation $Y_{it}(g_i, h_i(\mathbf{g}_{-i}))$, $ATT_0(g, t)$ is interpreted as the ATT when h_i is set to be zero for all units in group g. This aligns with the conventional definition of the ATT under SUTVA and is the estimand of interest in our paper. We can then define an aggregate ATT by $ATT_0 = \sum_{g,t} w_{gt} ATT_0(g, t)$, where w_{gt} is a weight chosen by the

econometrician (see, e.g., Callaway and Sant'Anna [2021]).[6]

We can also consider an alternative definition of the ATT that incorporates the spillover effects from other treated units, namely the units with group labels $g \leq t$:

$$ATT_S(g,t) \equiv \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g).$$

We refer to the difference $ATT_S(g,t) - ATT_0(g,t)$ as the average spillover effect on the treated:

$$AST(g,t) \equiv \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g).$$

Lastly, we define the corresponding average spillover effect for untreated units, which we refer to as the average spillover effect on the untreated[7]:

$$ASUT(g,t) \equiv \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g).$$

3 Intuition and Motivating Examples

To illustrate our setting, consider a panel data of six units ($i \in \{a, a', b, b', z, z'\}$) observed over three periods ($t = 1, 2, 3$), where units a, a' are treated in period 2, units b, b' are treated in period 3, and units z, z' are never-treated. The different effects of units' treatments on their outcomes are illustrated in Figure 1, focusing on periods 2 and 3. Solid arrows indicate the direct effect of a unit's own treatment, namely the integrand of ATT_0 (that is, $Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})$). Dotted arrows indicate the spillovers on the treated, i.e., the influence of unit j's treatment on unit i's outcome while unit i is treated, which add up to the integrand of AST (that is, $Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i})$). The dashed arrows represent the spillovers on

[6]Note that $ATT_0(g,t)$ is typically defined only for pairs (g,t) satisfying $t \geq g$. In this paper, we extend its definition to also include pairs satisfying $t < g$ with a trivial definition of $ATT_0(g,t) = 0$. We adopt this extension as it simplifies the notation in the proofs of our results.

[7]Note that the term "untreated units" refers to not-yet-treated units if $G_i < \infty$.

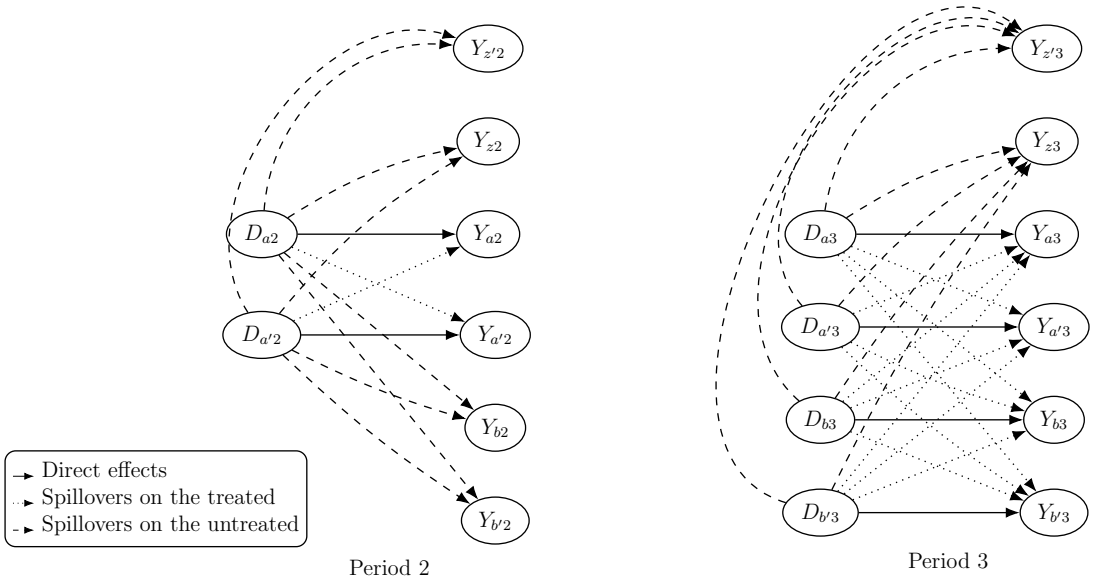


Figure 1: Illustration of the potential treatment and spillover paths.

the untreated, i.e., the influence of unit j's treatment on unit i's outcome while unit i is untreated, which add up to the integrand of $ASUT$ (that is, $Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{it}(\infty, \infty_{-i})$). Under SUTVA, only the direct effects represented by solid lines would exist.[8]

We now introduce two alternative scenarios (so-called Examples 1 and 2) involving spillover effects, which we will re-visit throughout the paper to motivate our assumptions and empirical application. In Example 1, the spillover takes the form of a diffusion effect, meaning that the direct effect and the spillover effects have the same sign. In Example 2, Gonzalez-Navarro [2013] motivates the possibility of displacement, where the direct effect and the spillover effects have opposite signs.

Example 1 (installation of a water treatment plant)**.** Consider a scenario where we are interested in the effect of introducing a water treatment plant on the health outcomes of villages situated along a river. Suppose the nearby villages a and a' are the first to adopt the plant. This adoption not only improves water quality in these villages but may also enhance

[8]For brevity of illustration, Figure 1 omits dynamic treatment effects. For instance, there are no arrows linking the treatment in period 2 to the outcomes in period 3.

the water quality of the not-yet-treated downstream villages, resulting in a spillover effect.

Example 2 (installation of stolen vehicle recovery devices)**.** Gonzalez-Navarro [2013] studied the effect of installing a stolen vehicle recovery device on car theft incidents. The introduction of this treatment was staggered across different states within a country and was limited to specific car models. In this scenario, the introduction of the device could generate spillover effects on other unprotected models within treated states, as well as on the same models in states that had not yet adopted the device. Gonzalez-Navarro [2013] reported an increase in thefts for the same models in states without the installed device.

There are two key challenges arising from these settings. First, untreated units might be affected by spillovers. Changes in the outcomes of untreated units over time would then be a combination of time trends and spillover effects. Especially in the case of displacement with staggered treatment adoption, spillover effects may intensify over time as more and more treated units spill on an increasingly narrower pool of untreated units. For instance, in Example 2, car theft could concentrate on only a few untreated states and models, causing large bias if used to estimate the time effects. Second, even if time fixed effects could be identified, separating the direct and spillover effects remains infeasible. However, isolating the direct effect is often of interest. For instance, when a unit decides whether to participate in a policy, its decision likely hinges on the direct effect, as spillovers from other units are beyond its control. Understanding the direct effect also provides insight into the mechanisms driving overall outcomes, which is essential for refining policies and optimizing resource allocation. Moreover, the sum of direct and spillover effects might have limited external validity, as spillovers depend on the specific treatment histories of other units, which may differ in alternative contexts. Policymakers and researchers concerned with scalability or transferability can benefit from isolating the direct effect to evaluate the intervention's core efficacy across diverse scenarios.

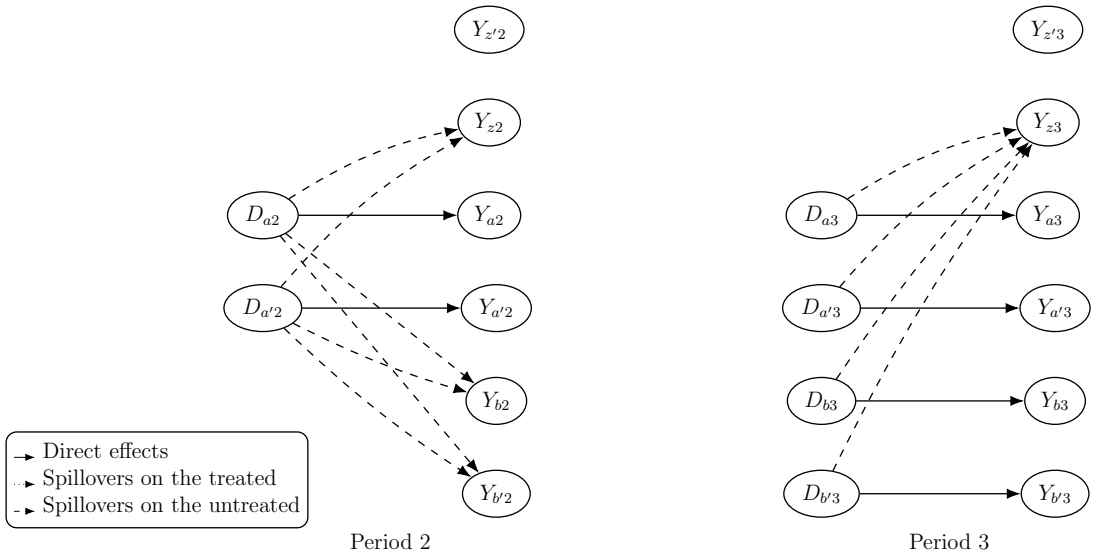


Figure 2: DAG under the key identification assumptions. There are no longer dotted arrows to treated outcomes and no arrows to $Y_{z'2}$ and $Y_{z'3}$.

Figure 2 visualizes our additional key assumptions, which will be formally introduced in Section 4, that resolve these issues and allow for identification of the direct effect. These are that units are not influenced by spillovers once treated, and that a subset of never-treated units remains unaffected by spillovers as well. Consequently, in this figure, dotted arrows and arrows to unit z' no longer appear. In what follows, we detail how these assumptions are likely to hold in empirical applications, illustrated through Examples 1 and 2.

Example 1 [continued]. Consider villages situated at the most upstream part of the river, none of which have water treatment plants. These upstream villages are not affected by the installation of water treatment plants in other villages along the river, since all other villages are downstream relative to them. Therefore, in this context, these upstream villages represent untreated units that are not subject to spillover effects.

Next, consider the village located furthest downstream, which initially does not have a water treatment plant. When an upstream village installs a plant, the downstream village experiences spillover effects, benefiting from improved water quality resulting from the upstream water treatment. However, once the downstream village installs its own water

treatment plant, the treatment status of the upstream village becomes irrelevant. The water quality in the downstream village is now only determined by its own treatment. Consequently, in this situation, treated units do not experience spillover effects.

A counterexample in which our assumption could possibly fail arises when the treatment itself generates harmful externalities instead, such as the opening of firms that emit toxic pollutants. In this case, pollution can travel across village boundaries (e.g., via prevailing winds, see Currie et al. [2015]), so that treated villages may continue to be affected by subsequent treatment adoption in nearby locations.

Example 2 [continued]. Consider states that are distant from all states where stolen vehicle recovery devices have been installed in specific car models. These states might be unaffected by spillover effects if behavioral responses to Lojack are geographically localized. Gonzalez-Navarro [2013] reports evidence consistent with the notion that geographical constraints limit displacement behavior. More generally, the same logic can be applied along a non-geographical dimension. If car models are imperfect substitutes in thieves' target choices, untreated car models that are dissimilar to the treated models may be plausibly unaffected by the installation of Lojack in treated models.

Next, consider a car model without the device, located in a state adjacent to one where the device has been installed. This car model may be subject to spillover effects if installing the device in the neighbouring state prompts thieves to redirect their targets toward unprotected vehicles in nearby areas. However, once the device is installed in these previously unprotected models, they no longer experience spillover effects, as thieves' attention turns to vehicles still lacking the device.[9]

[9]It might be argued that, as the coverage of states and car models with the protection device expands to become almost universal, thieves could eventually revert to targeting protected cars, violating the assumption. Even though this scenario might not be totally implausible, thieves would probably rather shift their focus to other, less protected assets or leave the criminal market entirely. For example, Cornish and Clarke [2017] point out that fitting steering column locks to all cars in West Germany during the 1960s resulted in a 60% reduction in car thefts without displacement, whereas introducing the locks solely on new cars in Great Britain merely displaced theft to older, unprotected vehicles. Moreover, such an almost universal adoption

A counterexample in which our assumption could possibly fail arises when the treatment is an increased police presence (see Di Tella and Schargrodsky [2004]), which, while effective, might not prevent all crimes. In this case, the treated areas may continue to be affected by subsequent treatment adoption in nearby locations.[10]

4 Identification

The discussion on the identification of $ATT_0(g,t)$ is structured into two steps. We first show that identifying $ATT_0(g,t)$ is equivalent to identifying the sum of the time effect and the spillover effect on the treated. The second step then introduces conditions that allow the identification of this sum. An implication of our assumptions is that it unifies the definitions of the ATT by implying that $ATT_0(g,t) = ATT_S(g,t)$.

We first present the necessary and sufficient condition for identifying $ATT_0(g,t)$ when spillovers are present.

Proposition 1. *Suppose that Assumptions 1 to 3 hold, and that all units are untreated at* $t = 1$. *Then, for every group* $g \in \mathcal{G}$ *such that* $2 \leq g < \infty$ *and time* $t \geq g$, *the parameter* $ATT_0(g,t)$ *is identified if and only if* $\delta_t + AST(g,t)$ *is identified, where* δ_t *is defined as in Remark 1.*

Proof. Refer to the Appendix for the proof of this proposition and of the results that follow. □

The intuition for Proposition 1 is as follows. The proof of Proposition 1 shows that,

of the protection device would be considered an extreme case and hard to evaluate, due to a very small set of control units.

[10] We thank an anonymous referee for this suggestion.

under Assumptions 1 to 3, for every (g,t) satisfying $t \geq g$:

$$\begin{aligned}
&\mathbb{E}(Y_{it}|G_i = g) = \mathbb{E}(Y_{it}(G_i, \mathbf{G}_{-i})|G_i = g) \\
&= \mathbb{E}(Y_{it}(\infty, \infty_{-i}) + [Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})] + [Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i})] \,|G_i = g) \\
&= \mathbb{E}(\alpha_i|G_i = g) + \delta_t + ATT_0(g,t) + AST(g,t),
\end{aligned}$$

where α_i and δ_t are defined as in Remark 1. Since $\mathbb{E}(\alpha_i|G_i = g)$ is identified by $\mathbb{E}(Y_{i1}|G_i = g)$, this implies that $ATT_0(g,t)$ is identified if the sum of δ_t (the time effect) and $AST(g,t)$ (the average spillover effect on the treated) is identified. In general, Assumptions 1 to 3 are not sufficient to identify this sum. Note that $AST(g,t) = 0$ in the absence of spillover effects, in which case $ATT_0(g,t)$ is identified if δ_t is identified.

In what follows, we propose two additional assumptions that enable identification of $ATT_0(g,t)$. We state the first assumption below.

Assumption 4 (no spillover effects on treated units)**.** *For every* (g,t) *such that* $t \geq g$*,*

$$\mathbb{E}(Y_{it}(g, \mathbf{G}_{-i})|G_i = g) = \mathbb{E}(Y_{it}(g, \infty_{-i})|G_i = g). \tag{4}$$

Remark 2. If the researcher is interested in $ATT_0(g,t)$ for a particular g, it is sufficient that Equation (4) holds for that particular g only. In this paper, we assume that Equation (4) holds for every g, since we will discuss aggregation of the $ATT_0(g,t)$ across g in later sections, which requires that Equation (4) holds for every g that is involved in the aggregation.

Remark 3. Assumption 4 is generally not testable. However, if one has prior knowledge about the sign of $ATT_0(g,t)$ and the possible sign of $AST(g,t)$ (if it exists), and if these signs are opposite, one can check whether the estimated sign of $ATT_0(g,t)$ is as expected. For example, if prior knowledge suggests that $ATT_0(g,t) < 0$ and $AST(g,t) \geq 0$, one can check whether the estimated $ATT_0(g,t)$ is negative, as otherwise it could imply that $AST(g,t) > 0$.

We use this remark in our application in Section 7.

Assumption 4 holds if $Y_{it}(G_i, \mathbf{G}_{-i}) = Y_{it}(G_i, \infty_{-i})$, implying that once a unit receives treatment, it is no longer influenced by spillover effects. This means the unit forfeits any spillovers it may have previously received and remains unaffected by spillovers from subsequently treated groups. We have previously discussed the plausibility of this assumption in Section 3, illustrated through Examples 1 and 2. Note that Assumption 4 is equivalent to $AST(g,t) = 0$, in which case $ATT_0(g,t) = ATT_S(g,t)$, unifying the definition of $ATT(g,t)$.

Next, we state the second assumption, which replaces Assumption 3.

Assumption 5 (existence of never-treated units unaffected by spillovers)**.** *There exists a positive mass of units within group $G_i = \infty$, where we define $H_i = 1$ for these units and define $H_i = 0$ for remaining units in $G_i = \infty$ and for all other units in $G_i \neq \infty$, such that*

$$\mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i})|G_i = \infty, H_i = 1) = \mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = \infty, H_i = 1), \tag{5}$$

and for all $g \in \mathcal{G}$,

$$\begin{aligned} &\mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = g) \\ &\qquad = \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1). \end{aligned} \tag{6}$$

Remark 4. Equation (5) states that there exists a subgroup of never-treated units that are not affected by spillovers, and Equation (6) states that parallel trends apply to said subgroup. We discuss a diagnostic test for checking the validity of these two equations in the Appendix.

Remark 5. Under Assumption 5, we can write, for every $g \in \mathcal{G}\backslash\{\infty\}$:

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = g) = \mathbb{E}(\alpha_i|G_i = g) + \delta_t,$$

and for $h \in \{0, 1\}$:

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = \infty, H_i = h) = \mathbb{E}(\alpha_i|G_i = \infty, H_i = h) + \delta_t,$$

where $\alpha_i = Y_{i1}(\infty, \infty_{-i})$, $\delta_t = \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1)$. Note that, with abuse of notation, we redefine δ_t here as a quantity that differs from the one defined in Remark 1.

Remark 6. Under Assumption 5, for every (g, t) satisfying $t \geq g$:

$$\begin{aligned}
&\mathbb{E}(Y_{it}|G_i = g) = \mathbb{E}(Y_{it}(G_i, \mathbf{G}_{-i})|G_i = g) \\
&= \mathbb{E}(Y_{it}(\infty, \infty_{-i}) + [Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})] + [Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i})] \,|G_i = g) \\
&= \mathbb{E}(\alpha_i|G_i = g) + \delta_t + ATT_0(g, t) + AST(g, t),
\end{aligned}$$

where α_i and δ_t are defined as in Remark 5. Note that δ_t defined as in Remark 5 is identified since

$$\delta_t = \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).$$

Remark 7. In practice, the researcher may take a conservative approach by setting $H_i = 1$ only for units strongly believed to satisfy Equation (5) and Equation (6), and $H_i = 0$ otherwise.

Remark 8. The researcher may also have knowledge of units believed to be unaffected by spillovers for $G_i \neq \infty$. In this case, we define $H_i = 1$ for these units and $H_i = 0$ for the remaining units in the same G_i group, and we could modify Assumptions 4 and 5 to hold

for each (G_i, H_i) group. For example, we could modify Equation (6) to

$$\begin{aligned}&\mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = g, H_i = h) \\ &\qquad = \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1).\end{aligned} \tag{7}$$

In the study by Gonzalez-Navarro [2013] described in Example 2, the author takes a conservative approach using only the never-treated states that are farthest from the treated ones as controls (Remark 7). Alternatively, in a scenario where spillover effects occur only among adjacent states, we could set $H_i = 1$ for all states not adjacent to any treated one until they become treated (Remark 8). In this case, the control group is not fixed but shrinks over t as more states adopt treatment.

Extending beyond Remark 8, there may be cases where different groups g' should be used for the parallel trend between period 1 and t, i.e., there exists $(g_2, \ldots, g_t) \in \mathcal{G}$ such that:

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i,t-1}(\infty, \infty_{-i})|G_i = g) = \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i,t-1}(\infty, \infty_{-i})|G_i = g_t, H_i = 1)$$

for each t and for all g. While this case also allows for identification and consistent estimation of ATT_0, the estimation procedure that will be introduced later becomes more complicated than under Assumption 5 or Remark 8, especially if the pattern of g_t is correlated with the potential outcomes. Since one of our focuses is to provide an estimation procedure that is straightforward to implement, we do not extend our setting to this case.

We conclude this section by showing that $ATT_0(g, t)$ is identified under these two assumptions, and that $ASUT(g, t)$ is identified under the latter assumption.

Theorem 1. *(a) Suppose that Assumptions 1, 2, 4 and 5 hold, and that all units are untreated at $t = 1$. Then, $ATT_0(g, t)$ is identified for every group $g \in \mathcal{G}$ such that $q \leq g < \infty$*

and time $t \geq g$. In particular,

$$ATT_0(g,t) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = g) - \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).$$

(b) If Assumptions 1, 2 and 5 hold and if all units are untreated at $t = 1$, $ASUT(g,t)$ is identified for every $g \in \mathcal{G}$ and $q \leq t < g$. In particular,

$$ASUT(g,t) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = g) - \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).$$

Theorem 1 shows that a single pre-treatment period ($t = 1$) is sufficient to identify ATT_0 and $ASUT$. However, for estimation, it is natural to utilize multiple pre-treatment periods ($t = 1, \ldots, q-1$) if they exist. In Section 5, we propose a general estimation procedure that exploits these multiple pre-treatment periods.

In addition, when there are multiple pre-treatment periods, one can modify Theorem 1 to identify ATT_0 and $ASUT$ under a weaker parallel trend assumption. Specifically, one can relax Equation (6) to:

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i,q-1}(\infty, \infty_{-i})|G_i = g) = \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{i,q-1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1),$$

where $q-1$ is the last pre-treatment period. Under this relaxation, the formula in Theorem 1 (a) modifies to:

$$ATT_0(g,t) = \mathbb{E}(Y_{it} - Y_{i,q-1}|G_i = g) - \mathbb{E}(Y_{it} - Y_{i,q-1}|G_i = \infty, H_i = 1).$$

By comparison, in the absence of spillovers, one may relax Assumption 3 to

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i,g-1}(\infty, \infty_{-i})|G_i = g) = \mathbb{E}(Y_{it}(\infty, \infty_{-i}) - Y_{i,g-1}(\infty, \infty_{-i})|G_i = \infty),$$

which states that the parallel trends hold relative to the group's last pre-treatment period $(g-1)$, rather than the last universal pre-treatment period $(q-1)$.

Note that, when Assumption 4 does not hold, the formula in Theorem 1 (a) modifies to:

$$ATT_0(g,t) + AST(g,t) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = g) - \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).$$

Therefore, in the absence of Assumption 4, the right-hand side of the formula identifies $ATT_0(g,t) + AST(g,t)$, the sum of the direct and spillover effects, which equals $ATT_S(g,t)$ defined in Section 2. Consequently, the estimation procedure discussed in the next section consistently estimates $ATT_S(g,t)$ rather than $ATT_0(g,t)$.

In this circumstance, if it is reasonable to assume that the magnitude of the average spillover effect on the treated is no greater than that of the average spillover effect on the untreated, i.e., $|AST(g,t)| \leq |ASUT(g,g-1)|$ for all $t \geq g > q$, the lower and upper bounds of the $ATT_0(g,t)$ can be constructed in the spirit of Rambachan and Roth [2023]:

$$\Delta_Y - |ASUT(g,g-1)| \leq ATT_0(g,t) \leq \Delta_Y + |ASUT(g,g-1)|,$$

where $\Delta_Y \equiv \mathbb{E}(Y_{it} - Y_{i1} \mid G_i = g) - \mathbb{E}(Y_{it} - Y_{i1} \mid G_i = \infty, H_i = 1)$.

5 Estimation and Inference

In this section, we discuss estimation and inference of $ATT_0(g,t)$ and $ASUT(g,t)$. Consider a balanced panel of T periods, where all units are untreated at $t = 1$. For estimation, it is useful to introduce a binary variable S_{it} that indicates whether an observation (i,t) could be subject to spillover effects. To define this variable, recall that an observation is potentially influenced by spillovers under the following three conditions. First, there exists a treated unit (i.e., $t \geq q$, where q is the first period that any unit enters treatment).

Second, the observation is not treated (i.e., $D_{it} = 0$), as otherwise treated observations are not influenced by spillover effects by Assumption 4. Lastly, the observation has $H_i = 0$, as otherwise untreated units with $H_i = 1$ are considered unaffected by spillover effects by Assumption 5. Therefore, we define S_{it} as a binary variable that is equal to 1 if the following three conditions hold: (i) $t \geq q$, (ii) $D_{it} = 0$, and (iii) $H_i = 0$.

We first discuss the estimation in the case where units with $H_i = 1$ exist only within group $G_i = \infty$. It is useful to introduce an extended group label that partitions $G_i = \infty$ into $(G_i = \infty, H_i = 1)$ and $(G_i = \infty, H_i = 0)$. Define $\tilde{\mathcal{G}}$ as the support of $(G_i, H_i) \in \mathcal{G} \times \{0, 1\}$. For example, if $\mathcal{G} = \{q, q+1, \ldots, T, \infty\}$, then $\tilde{\mathcal{G}} = \{(q, 0), (q+1, 0), \ldots, (T, 0), (\infty, 0), (\infty, 1)\}$.

We propose the following extension of Wooldridge [2025] as the estimation procedure. We estimate a linear regression model where Y_{it} is the outcome variable, and the regressors are indicators of (G_i, H_i) (the "extended group fixed effects"), indicators of t (the "time fixed effects"), interactions between D_{it} and indicators of (G_i, H_i, t), and interactions between S_{it} and indicators of (G_i, H_i, t). In other words, we estimate the linear regression model:

$$\begin{aligned} Y_{it} = \alpha_{G_i H_i} + \delta_t + \sum_{(g',h') \in \tilde{\mathcal{G}}} \sum_{t'} \beta_{g'h't'} \cdot \mathbf{1}((G_i, H_i, t) = (g', h', t')) \cdot D_{it} \\ + \sum_{(g',h') \in \tilde{\mathcal{G}}} \sum_{t'} \gamma_{g'h't'} \cdot \mathbf{1}((G_i, H_i, t) = (g', h', t')) \cdot S_{it} + \varepsilon_{it}. \end{aligned} \tag{8}$$

Since this equation contains multicollinear terms, it can also be written as

$$\begin{aligned} Y_{it} = \alpha_{G_i H_i} + \delta_t + \sum_{g' \in \mathcal{G} \setminus \{\infty\}} \sum_{t'=g'}^{T} \beta_{g't'} \cdot \mathbf{1}((G_i, t) = (g', t')) \cdot D_{it} \\ + \sum_{(g',h') \in \tilde{\mathcal{G}} \setminus \{(\infty,1)\}} \sum_{t'=q}^{\min\{g'-1, T\}} \gamma_{g'h't'} \cdot \mathbf{1}((G_i, H_i, t) = (g', h', t')) \cdot S_{it} + \varepsilon_{it}. \end{aligned} \tag{9}$$

Note that Equation (9) involves the *group*-level coefficients $(\alpha_{gh}, \beta_{gt})$, as opposed to the

unit-level coefficients (α_i, β_{it}).[11] The following result shows that, despite this simplification, the population regression of Equation (9) correctly identifies $ATT_0(g,t)$ and $ASUT(g,t)$.

Theorem 2. *Suppose that the assumptions of Theorem 1 hold. Consider the population regression of Equation (9). Then*

$$\beta_{gt} = \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i}) | G_i = g),$$
$$\gamma_{g,0,t} = \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{it}(\infty, \infty_{-i}) | G_i = g, H_i = 0).$$

Note that Theorem 2 implies that $\beta_{gt} = ATT_0(g,t)$ and $\gamma_{g,0,t} = ASUT(g,t)$ for $2 \leq g < \infty$. In addition, $\gamma_{\infty,0,t}$ equals $ASUT$ for the subgroup $(G_i = \infty, H_i = 0)$. Then, $ASUT(\infty, t)$ can be calculated as $ASUT(\infty,t) = w_{\infty,0,t} \cdot \gamma_{\infty,0,t} + w_{\infty,1,t} \cdot 0 = w_{\infty,0,t}\gamma_{\infty,0,t}$, where $w_{\infty,0,t}$ and $w_{\infty,1,t}$ are the weights for the subgroups $(G_i = \infty, H_i = 0)$ and $(G_i = \infty, H_i = 1)$, respectively. Note that the spillover effect on $(G_i = \infty, H_i = 1)$ is zero.

Theorem 2 yields a simple procedure for estimation and inference of $ATT_0(g,t)$ (or $ASUT(g,t)$), since $\widehat{\beta}_{gt}$ (or $\widehat{\gamma}_{g,0,t}$) and its standard error can be easily obtained by implementing Equation (9) using standard statistical software. Moreover, estimation and inference of an aggregate ATT_0 is also straightforward. The estimate of an aggregate ATT_0 is given by $\sum_{g,t} w_{gt}\widehat{\beta}_{gt}$, and its standard error is given by

$$\text{Var}\left(\sum_{g,t} w_{gt}\widehat{\beta}_{gt}\right) = \sum_{g,t}\sum_{g',t'} w_{gt}w_{g't'}\text{Cov}(\widehat{\beta}_{gt}, \widehat{\beta}_{g't'}),$$

where $\text{Cov}(\widehat{\beta}_{gt}, \widehat{\beta}_{g't'})$ is the covariance between the two individual $ATT_0(g,t)$ estimators, namely $\widehat{\beta}_{gt}$ and $\widehat{\beta}_{g't'}$. In order to obtain a valid standard error of an aggregate ATT_0, one must learn about all the covariances between the individual $ATT_0(g,t)$ estimators, which may not be a straightforward task. In our estimation procedure, these covariances can be

[11] This will become important when comparing our estimation procedure with imputation-based procedures later in this section, and when discussing estimation in the nonlinear case in Section 6.

obtained conveniently from standard statistical software, e.g., via e(V) in Stata.

It is worth noting that Equation (9) is numerically equivalent to the following extension of the imputation procedure of Borusyak et al. [2024]:

1. Estimate $Y_{it} = \alpha_i + \delta_t + \varepsilon_{it}$, using observations (i,t) such that $D_{it} = 0$ and $S_{it} = 0$.

2. Let $\widehat{\alpha}_i$ and $\widehat{\delta}_t$ be the estimates of α_i and δ_t, and impute the baseline outcome for unit i at time t as $\widehat{Y}_{it}(\infty, \infty_{-i}) = \widehat{\alpha}_i + \widehat{\delta}_t$.

3. For each unit i treated at time t, compute $\widehat{\beta}_{it}^{imp} \equiv Y_{it} - \widehat{Y}_{it}(\infty, \infty_{-i})$.

4. For a treated group g at time $t \geq g$, estimate $ATT_0(g,t)$ by $\widehat{\beta}_{gt}^{imp} \equiv (1/N_g) \sum_{i=1}^{N_g} \widehat{\beta}_{it}^{imp}$, where $N_g = \sum_{i=1}^{N} \mathbf{1}(G_i = g)$.

While it can be shown that $\widehat{\beta}_{gt}^{imp}$ equals $\widehat{\beta}_{gt}$ in Equation (9), Borusyak et al. [2024] highlight the challenge in estimating the standard error of the imputation estimate, which arises from computing $\widehat{\beta}_{it}^{imp}$ for each i and t. In comparison, the use of group-level coefficients in Equation (9) eases the calculation of standard errors and covariances of the $\widehat{\beta}_{gt}$s, which can be done conveniently using standard statistical software.

Next, we consider the case where there are also units with $H_i = 1$ under $G_i \neq \infty$. Recall that the extended group label (G_i, H_i) partitions each group $G_i = g$ into subgroups $(G_i = g, H_i = 1)$ and $(G_i = g, H_i = 0)$. We define ATT_0 and $ASUT$ for these subgroups as follows:

$$ATT_0(g,h,t) = \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i}) | G_i = g, H_i = h),$$

$$ASUT(g,h,t) = \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{it}(\infty, \infty_{-i}) | G_i = g, H_i = h).$$

We introduce this new definition because the ATT_0 (or $ASUT$) for the subgroups $(G_i = g, H_i = 0)$ and $(G_i = g, H_i = 1)$ may differ.[12] The aggregate ATT_0 can then be defined as $ATT_0 = \sum_{g,h,t} w_{ght} ATT_0(g,h,t)$, where w_{ght} is a weight chosen by the econometrician. The

[12] In the previous case, where units with $H_i = 1$ existed only within group $G_i = \infty$, there were no units with $(G_i = g, H_i = 1)$ when $g \neq \infty$, so $ATT_0(g,t) = ATT_0(g,0,t)$.

same estimation procedure described earlier applies to this case as well, where we run the full regression as described in Equation (8), and the coefficient β_{ght} equals $\beta_{ght} = ATT_0(g,h,t)$ and γ_{ght} equals $ASUT(g,h,t)$.

Lastly, we illustrate an extension of our estimation procedure that allows for additional covariates, in the spirit of Wooldridge [2023]. Specifically, for some covariate vector X_{it}, one may assume that Assumption 2 holds conditional on X_{it} and extend Assumption 5 and its characterization in Remark 5 to a form similar to Assumptions CIPT and CIPTS in Wooldridge [2023]. To illustrate, a stylized specification of this type is

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = g, H_i = h, X_i) = \alpha_{gh} + \delta_t + \gamma' X_{it},$$

where $X_i = (X_{i1}, \ldots, X_{iT})'$. Note that X_{it} should also be assumed to be unaffected by treatment or spillovers, as discussed in Caetano et al. [2026]. Then, under the assumption that the treatment and spillover effects do not vary with X_i, the estimation procedure discussed in this section can be extended to incorporate covariates as additional regressors. For a comprehensive development of this approach, see Sections 2.2 and 3 of Wooldridge [2023].

6 Extension to Nonlinear DiD Models

In this section, we extend our approach to the case where Y_{it} is a limited dependent variable, such as binary or count variables, for which the linear parallel trends assumption may not hold. This extension contributes to the literature on nonlinear DiD models [Wooldridge, 2023], expanding the applicability of our results to a wider array of empirical applications.

We introduce the following parallel trends assumption in the context of nonlinear models, following Wooldridge [2023]. Let $F : \mathbb{R} \mapsto \mathbb{R}$ be a monotonically increasing function.

Assumption 3' (parallel trends, nonlinear model)**.** *For every group g at time t,*

$$\begin{aligned} &F\left(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=g)\right)-F\left(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=g)\right) \\ &=F\left(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty)\right)-F\left(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty)\right) \end{aligned}. \tag{10}$$

Common choices of F are canonical link functions, corresponding to nonlinear regressions that are frequently used in empirical applications. For instance, if Y_{it} is binary, F can be set as the inverse of the Gaussian CDF, corresponding to probit regression, or the inverse of the Logistic function, corresponding to logit regression. If Y_{it} is a count variable, F can be set as the log function, which corresponds to Poisson regression.

By replicating the arguments in Proposition 1 and Theorem 1, the following corollaries show that $ATT_0(g,t)$ is identified under assumptions similar to those in Theorem 1. In doing so, we adapt Assumption 5 to the case of nonlinear models, replacing Assumption 3'.

Assumption 5' (existence of never-treated units unaffected by spillovers)**.** *There exists a positive mass of units within group $G_i=\infty$, where we define $H_i=1$ for these units and define $H_i=0$ for remaining units in $G_i=\infty$ and for all other units in $G_i\neq\infty$, such that*

$$\mathbb{E}(Y_{it}(\infty,\mathbf{G}_{-i})|G_i=\infty,H_i=1)=\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty,H_i=1), \tag{11}$$

and for $g\in\mathcal{G}\backslash\{\infty\}$,

$$\begin{aligned} &F(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=g))-F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=g)) \\ &\quad=F(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty,H_i=1))-F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty,H_i=1)). \end{aligned} \tag{12}$$

and

$$F(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty,H_i=0))-F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty,H_i=0))$$
$$=F(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty,H_i=1))-F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty,H_i=1)). \tag{13}$$

Assumptions 3 and 5 are special cases of Assumptions 3' and 5' where F is an identity function, i.e., $F(x)=x$.

Remark 9. Under Assumption 5', we can write, for every $g\in\mathcal{G}\backslash\{\infty\}$:

$$\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=g)=F^{-1}(\alpha_g+\delta_t), \tag{14}$$

and for $H_i\in\{0,1\}$:

$$\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty,H_i=h)=F^{-1}(\alpha_{\infty,h}+\delta_t), \tag{15}$$

where $\alpha_g=F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=g))$, $\alpha_{\infty,h}=F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty,H_i=h))$ and $\delta_t=F(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty,H_i=1))-F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty,H_i=1))$.

Corollary 1. *Suppose that Assumptions 1, 2, and 3' hold, and that all units are untreated at $t=1$. Then, for every group $g\in\mathcal{G}$ such that $2\leq g<\infty$ and time $t\geq g$, the $ATT_0(g,t)$ is identified if and only if $F^{-1}(F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=g))+[F(\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i=\infty))-F(\mathbb{E}(Y_{i1}(\infty,\infty_{-i})|G_i=\infty))])+AST(g,t)$ is identified.*

Corollary 2. *(a) Suppose that Assumptions 1, 2, 4, and 5' hold, and that all units are untreated at $t=1$. Then, $ATT_0(g,t)$ is identified for every group $g\in\mathcal{G}$ such that $2\leq g<\infty$ and time $t\geq g$. In particular,*

$$ATT_0(g,t)=\mathbb{E}(Y_{it}|G_i=g)-F^{-1}\Big[F(\mathbb{E}(Y_{i1}|G_i=g))+$$
$$F(\mathbb{E}(Y_{it}|G_i=\infty,H_i=1))-F(\mathbb{E}(Y_{i1}|G_i=\infty,H_i=1))\Big].$$

(b) If Assumptions 1, 2, and 5' hold and if all units are untreated at $t = 1$, $ASUT(g,t)$ is identified for every $g \in \mathcal{G}\backslash\{\infty\}$ and $q \leq t < g$. In particular,

$$ASUT(g,t) = \mathbb{E}(Y_{it}|G_i = g) - F^{-1}\Big[F(\mathbb{E}(Y_{i1}|G_i = g)) + F(\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)) - F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1))\Big].$$

Moreover, ASUT for the subgroup $(G_i = \infty, H_i = 0)$ is also identified. In particular,

$$ASUT(\infty,0,t) = \mathbb{E}(Y_{it}|G_i = \infty, H_i = 0) - F^{-1}\Big[F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 0)) + F(\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)) - F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1))\Big].$$

For example, when F is the log function, $ATT_0(g,t)$ is identified by, for $t \geq g$,

$$ATT_0(g,t) = \mathbb{E}(Y_{it}|G_i = g) - \mathbb{E}(Y_{i1}|G_i = g)\frac{\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)}{\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1)}.$$

Estimation and inference are also similar to Section 5. To be concrete, we discuss details on estimation and inference in the case where F is a log function, corresponding to Poisson regression for count data, which we use in our application in the next section. The estimation and inference procedures can be similarly applied for any canonical link function F.

Let S_{it} be defined as in the previous section, and consider a balanced panel of T periods where all units are untreated at $t = 1$. In the case of count data, the average treatment effect in terms of percentage changes is also often reported:

$$ATTP_0(g,t) = \frac{ATT_0(g,t)}{\mathbb{E}(Y_{it}(\infty,\infty_{-i})|G_i = g)},$$

which can be aggregated to define an $ATTP_0 \equiv \sum_{g,t} w_{gt} ATTP_0(g,t)$.

Consider the case where units with $H_i = 1$ are all within group $G_i = \infty$. We use the following simple estimation procedure that involves a parsimonious generalized linear model.

1. Estimate the Poisson regression model where Y_{it} is the outcome variable, and the regressors are indicators of (G_i, H_i) (the "extended group fixed effects"), indicators of t (the "time fixed effects"), interactions between D_{it} and indicators of (G_i, H_i, t), and interactions between S_{it} and indicators of (G_i, H_i, t). In other words, we estimate the Poisson regression model:

$$\begin{aligned}\ln \mathbb{E}(Y_{it}|\mathbf{X}_{it}) =& \alpha_{G_i H_i} + \delta_t + \sum_{g' \in \mathcal{G}\setminus\{\infty\}} \sum_{t'=g'}^{T} \beta_{g't'} \cdot \mathbf{1}((G_i, t) = (g', t')) \cdot D_{it} \\ &+ \sum_{(g',h') \in \tilde{\mathcal{G}}\setminus\{(\infty,1)\}} \sum_{t'=q}^{\min\{g'-1,T\}} \gamma_{g'h't'} \cdot \mathbf{1}((G_i, H_i, t) = (g', h', t')) \cdot S_{it},\end{aligned} \tag{16}$$

   where $\mathbf{X}_{it}$ represents the vector of regressors. Let $\widehat{\alpha}_{gh}$, $\widehat{\delta}_t$, and $\widehat{\beta}_{gt}$ be the estimates of α_{gh}, δ_t, and β_{gt} from this model, respectively.

2. For $2 \leq g < \infty$ and $t \geq g$, estimate $ATT_0(g,t)$ by:

$$\widehat{ATT}_0(g,t) = \exp\{\widehat{\alpha}_{g,0} + \widehat{\delta}_t + \widehat{\beta}_{gt}\} - \exp\{\widehat{\alpha}_{g,0} + \widehat{\delta}_t\},$$

   or estimate $ATTP_0(g,t)$ by $\widehat{ATTP}_0(g,t) = \exp\{\widehat{\beta}_{gt}\} - 1$. Similarly, for $2 \leq g < \infty$ and $q \leq t < g$, estimate $ASUT(g,t)$ by

$$\widehat{ASUT}(g,t) = \exp\{\widehat{\alpha}_{g,0} + \widehat{\delta}_t + \widehat{\gamma}_{g,0,t}\} - \exp\{\widehat{\alpha}_{g,0} + \widehat{\delta}_t\}.$$

For a generic canonical link function F, $\widehat{ATT}_0(g,t) = F^{-1}(\widehat{\alpha}_{g,0} + \widehat{\delta}_t + \widehat{\beta}_{gt}) - F^{-1}(\widehat{\alpha}_{g,0} + \widehat{\delta}_t)$, where $(\widehat{\alpha}_{g,0}, \widehat{\beta}_{gt}, \widehat{\delta}_t)$ are estimates from the generalized linear model corresponding to F, such as probit or logit. A similar argument applies to $\widehat{ASUT}(g,t)$.

The validity of the population regression of Equation (16) can be shown by replicating the arguments in Theorem 2, and we omit the proof here. The estimation and inference

of $\widehat{ATT}_0(g,t)$ can then be carried out by implementing Equation (16) using any standard statistical software that runs Poisson regressions.

Note that most statistical software that run Poisson regressions calculate the standard errors of $(\widehat{\alpha}_{gh}, \widehat{\delta}_t, \widehat{\beta}_{gt})$ using the maximum likelihood. This assumes that the distribution of $Y_{it}(\infty, \infty_{-i})$ follows a Poisson distribution, as opposed to only specifying its mean as in Assumption 5', ruling out heteroskedasticity. To accommodate heteroskedasticity, standard errors can instead be derived using the quasi-maximum likelihood estimation (QMLE) method. Specifically, let θ be the vector of all coefficients in the Poisson regression (i.e., all of α_{gh}, δ_t, β_{gt}, and γ_{ght}), $\hat{\theta}$ be their maximum likelihood estimates (i.e., all of $\widehat{\alpha}_{gh}$, $\widehat{\delta}_t$, $\widehat{\beta}_{gt}$, and $\widehat{\gamma}_{ght}$), and $\mathbf{X}_{it}$ be the vector of all regressors. Let $\{\Lambda^c\}_{c=1}^C$ be the partition of units according to which the units are clustered. Define

$$\mathcal{S} = \sum_{c=1}^{C} \left[\sum_{i \in \Lambda^c} \sum_{t=1}^{T} \mathbf{X}_{it}(Y_{it} - \widehat{Y}_{it}) \right] \left[\sum_{i \in \Lambda^c} \sum_{t=1}^{T} \mathbf{X}_{it}(Y_{it} - \widehat{Y}_{it}) \right]'$$

as the clustered outer product of the score function, where $\widehat{Y}_{it} = \exp\{X_{it}'\hat{\theta}\}$ is the fitted value of Y_{it} in the Poisson regression.[13] In addition, define $\mathcal{H} = \sum_{c=1}^{C} \sum_{i \in \Lambda^c} \sum_{t=1}^{T} \mathbf{X}_{it}\mathbf{X}_{it}'\widehat{Y}_{it}$ as the negative Hessian. Then, the variance matrix of $\hat{\theta}$ is given by $\widehat{\text{Var}}(\hat{\theta}) = \mathcal{H}^{-1}\mathcal{S}\mathcal{H}^{-1}$.

In practice, $\widehat{\text{Var}}(\hat{\theta})$ is available in most software packages, e.g., via vce(cluster clustvar) in Stata. $\widehat{\text{Var}}(\hat{\theta})$ can then be used to compute the standard errors of $\widehat{ATT}_0$ and $\widehat{ATTP}_0$ via the delta method. Specifically, we write $\widehat{ATT}_0(g,t)$ as a function of $\hat{\theta}$, denoted by $f(\hat{\theta})$:

$$\widehat{ATT}_0(g,t) = \exp\{\widehat{\alpha}_{g,0} + \widehat{\delta}_t + \widehat{\beta}_{gt}\} - \exp\{\widehat{\alpha}_{g,0} + \widehat{\delta}_t\} \equiv f(\hat{\theta}).$$

Then, by the delta method,

$$\widehat{\text{Var}}\left(\widehat{ATT}_0(g,t)\right) \approx \nabla f(\hat{\theta})' \widehat{\text{Var}}(\hat{\theta}) \nabla f(\hat{\theta}),$$

[13] We abuse notation and let $\widehat{Y}_{it}$ represent a different object from the linear case.

where ∇f is the gradient of f. The procedure for $\widehat{ATTP}_0$ is similar, with a different form of f. The approach for a generic canonical link function F is also similar, with the score and the Hessian computed according to the nonlinear regression model that corresponds to F.

7 Application to Auto Theft Prevention Policy

In this section, we apply our method to revisit the findings of Gonzalez-Navarro [2013], who uses a nonlinear model to study the effects of installing an auto theft prevention device known as Lojack. Lojack is a compact device installed in vehicles that allows stolen cars to be tracked and recovered.

The policy was implemented in Mexico through an exclusive agreement between the Ford Motor Company and the Lojack company. Initially, Lojack was introduced for the Ford Windstar model in the state of Jalisco, beginning from its 2001 car models. Subsequently, its installation expanded to other *model* $\times$ *state* combinations, eventually covering 32 *model* $\times$ *state* combinations by 2004. The dataset of Gonzalez-Navarro [2013] provides information on the number of car thefts for each *model* $\times$ *state* $\times$ *vintage* (the car model's year) combination for each calendar year. For our analysis, we use the indices m, s, v, and t to denote car model, state, vintage, and the calendar year of the auto theft, respectively.

Gonzalez-Navarro [2013] points to two possible dimensions of spillover effects following the introduction of Lojack. The first is a model-level, non-geographical spillover within Lojack-installed states. Because Lojack is installed only in selected car models, thieves may respond by shifting toward models that are not equipped with Lojack within the same state. The second is a geographical spillover to nearby states without Lojack. Thieves, particularly those specializing in models that become equipped with Lojack, may relocate their activity to states where those same models remain unprotected. These two spillover dimensions need not operate independently. If the introduction of Lojack induces some thieves to move their

activity to other states, the resulting reduction in theft opportunities or criminal activity within Lojack-installed states may also reduce thefts of models that are not themselves equipped with Lojack.

Because of the potential for such spillovers, Gonzalez-Navarro [2013] relies only on time-series variation for identification, illustrating the challenge in extending the DiD framework to spillovers. Nevertheless, as a robustness check, Gonzalez-Navarro [2013] also estimates a DiD model while attempting to control for spillover effects, but without accounting for the staggered adoption design. In this section, we apply our method to revisit this study and estimate the treatment effect across various combinations of groups and time periods, thereby revealing the heterogeneous effects of Lojack installation.

Once Lojack was installed in a particular combination of car model, state, and vintage, it continued to be installed in all subsequent vintages of that model in the same state. This setup allows us to treat the situation as a staggered adoption design, where the unit of analysis is defined as the combination of *model* (m) $\times$ *state* (s) $\times$ *age* (a), where *age* refers to the number of years elapsed since the car's model year, calculated as $a = t - v$. Under this framework, our analysis is based on a balanced panel subset derived from the original dataset, consisting of 1152 units observed annually from 1999 to 2004.

Let D_{msat} be the binary treatment indicator for a unit (m, s, a) at time t. To illustrate, consider the Ford Windstar model in the state of Jalisco. For this unit, Lojack has been installed in all newly released ($age = 0$) vehicles beginning from 2001. Thus, for a Ford Windstar in Jalisco with $age = 0$, we have $D_{Windstar,Jalisco,0,t} = 1$ for every $t \geq 2001$.

To apply our method, we need to check the credibility of Assumptions 4 and 5' in this context. Assumption 4 states that once a *model* $\times$ *state* $\times$ *age* unit is protected by Lojack, the subsequent expansion of Lojack to other units does not affect the unit's theft count. Generally, when Lojack is installed in certain units, we can expect that thieves targeting those models will shift their focus towards vehicles without Lojack protection, rather than

those already with Lojack. Thus, it is plausible that once a unit is itself protected by Lojack, subsequent installations elsewhere do not materially affect its theft count. Assumption 5' requires the existence of units that are unaffected by spillover effects. The choice of these units, denoted by $H_i = 1$, requires researcher discretion and need not be based on geographical distance. For example, one could define spillover-free units using geographical distance, similarity across car models, or a combination of the two. In our application, we follow Gonzalez-Navarro [2013] and take a conservative geographical approach: we define $H_{msa} = 1$ for units belonging to states in the farthest distance tercile from the Lojack-installed states, regardless of car model. We then define $S_{msat} = 1$ if $t \geq 2001$, $D_{msat} = 0$, and $H_{msa} = 0$. Consequently, the set of untreated units potentially exposed to spillovers includes both unprotected models within Lojack-installed states and units in nearby states, allowing spillovers to operate along both non-geographical and geographical dimensions. Later in this section, we also discuss results for the diagnostic tests mentioned in Remarks 3 and 4, which support the credibility of Assumptions 4 and 5'.

Let Y_{msat} be the number of auto thefts for a *model* $\times$ *state* $\times$ *age* unit that occurred in a given calendar year t. We consider two kinds of empirical models for this outcome, namely the linear and Poisson parallel trends:

$$(i) \qquad \mathbb{E}(Y_{msat}(\infty, \infty_{-(msa)})|G_{msa} = g, H_{msa} = h) = \alpha_{gh} + \delta_t,$$
$$(ii) \quad \ln \mathbb{E}(Y_{msat}(\infty, \infty_{-(msa)})|G_{msa} = g, H_{msa} = h) = \alpha_{gh} + \delta_t,$$

where (i) aligns with Assumption 5, where the combination (m, s, a) plays the role of i, and (ii) aligns with Assumption 5'. In our application, (ii) is particularly suitable because Y_{msat} is a count variable with a high frequency of zeros.

Define $\overline{\mathcal{G}} = \{2001, 2002, 2003, 2004\}$ as the set of group labels for treated units, namely the periods at which units enter treatment. Let N_g be the number of units in group $g \in \overline{\mathcal{G}}$ within the dataset, and let $\overline{N} \equiv \sum_{g=2001}^{2004} \sum_{t=g}^{2004} N_g = \sum_{g=2001}^{2004} (2005 - g) N_g$ be the total

	Linear				Poisson			
	Estimate	SE(st)	SE(id)	Reduction	Estimate	SE(st)	SE(id)	Reduction
ATT_0	-9.258	5.169	2.545	-74.1%	-8.499	5.648	2.868	-71.7%
ATT_0^0	-7.131	4.257	2.043	-61.6%	-5.967	4.481	2.298	-57.3%
ATT_0^1	-9.957	5.465	2.745	-83.2%	-9.519	6.121	3.180	-82.5%
ATT_0^2	-19.585	10.973	10.970	-90.2%	-19.222	12.166	11.293	-89.2%

Table 1: Estimates of the aggregate ATT_0. The standard errors are clustered at the state (SE(st)) and the unit (SE(id)) levels. The "Reduction" column stands for the reduction rate, which is calculated using the formula for computing $ATTP_0$.

number of treated observations in the dataset. We estimate the following aggregate ATT_0:

$$ATT_0 = \sum_{g=2001}^{2004} \sum_{t=g}^{2004} \frac{N_g}{\overline{N}} ATT_0(g,t),$$
$$ATT_0^k = \sum_{g=2001}^{2004-k} \frac{N_g}{N_{2001} + \cdots + N_{2004-k}} ATT_0(g, g+k), \qquad k = 0, 1, 2.$$

In the above definitions, ATT_0 measures the overall effect of Lojack installation, computed as the weighted average of all $ATT_0(g,t)$ values across g and t. The ATT_0^k values, on the other hand, represent the weighted average of ATT_0 for the k-th year after installation of Lojack, measuring the temporal effects. For example, ATT_0^0 represents the immediate effect in the same year as the Lojack installation, ATT_0^1 represents the effect one year post-installation, and so forth.

Table 1 presents the estimated ATT_0 values obtained from both linear and Poisson model specifications, with standard errors clustered at the state and the unit levels. Table 1 reveals a notable average reduction in thefts of 74.1% for the linear model and 71.7% for the Poisson model, highlighting Lojack's substantial deterrent effect. Moreover, the results from both models indicate that the rate of theft reduction becomes more pronounced over time. This highlights the increasing effectiveness of Lojack in preventing auto thefts over time.

For comparison, we also report the estimated ATT_0 from two misspecified models. First,

	TWFE (linear)		BW (linear)	
	Estimate	Reduction	Estimate	Reduction
ATT_0	-8.320	-70.0%	-7.834	-70.1%
ATT_0^0	N/A		-5.637	-55.4%
ATT_0^1	N/A		-8.453	-81.0%
ATT_0^2	N/A		-19.138	-87.8%
	TWFE (Poisson)		BW (Poisson)	
	Estimate	Reduction	Estimate	Reduction
ATT_0	-6.940	-66.1%	-5.851	-63.9%
ATT_0^0	N/A		-3.957	-46.3%
ATT_0^1	N/A		-6.489	-76.7%
ATT_0^2	N/A		-14.574	-86.4%

Table 2: Estimates of the aggregate ATT_0 using the TWFE specification that incorporates spillover effects but does not account for staggered adoption (the "TWFE" columns), and the specification of Borusyak et al. [2024] and Wooldridge [2025] that does not account for spillover effects (the "BW" columns), for each of linear and Poisson specifications. The "Reduction" columns stand for the reduction rate, which is calculated using the formula for computing $ATTP_0$.

we consider the standard TWFE specification that incorporates spillover effects but overlooks the staggered nature of the treatment. Second, we consider the specification of Borusyak et al. [2024] and Wooldridge [2025] that accounts for staggered adoption but does not include spillover effects. Results from these models are presented in Table 2. Relative to the correctly specified estimates in Table 1, both sets of misspecified estimates imply a smaller reduction in car theft.

Next, we report the estimates of the spillover effects and the diagnostic test results. First, we estimate the calendar-time-aggregated $ASUT$ for each $t = 2001, 2002, 2003, 2004$:

$$ASUT_t = \frac{\sum_{g=t+1}^{2004} N_g ASUT(g,0,t) + N_{\infty,0} ASUT(\infty,0,t)}{\sum_{g=t+1}^{2004} N_g + N_{\infty,0}},$$

where N_g is the number of units in group g. By construction, $ASUT_t$ averages the spillover effects across all untreated units that are potentially exposed to spillovers at time t. Im-

portantly, this aggregate need not correspond to a single dimension of exposure. In our application, potentially exposed untreated units fall into two groups: (i) untreated units in Lojack-installed states, which may be affected through model-level, non-geographical spillovers, and (ii) untreated units in nearby states, which may be affected through geographical spillovers. Thus, the overall $ASUT_t$ aggregates spillover effects operating through both dimensions.

To examine these dimensions separately, we also estimate dimension-specific spillover effects, assuming that the relevant parallel trends conditions of the type in Assumptions 5 and 5' hold for the corresponding subgroups. Estimating $ASUT_t$ on the subsample that excludes units in group (i) identifies the spillover effect for units in group (ii). Conversely, excluding units in group (ii) identifies the spillover effect for units in group (i). In each case, $ASUT_t$ is calculated using only the remaining untreated units that are potentially exposed to spillovers. Alternatively, following the placebo-test approach discussed in the Appendix, one can assign distinct labels to groups (i) and (ii) and then estimate $ASUT_t$ separately for each group.

Table 3 presents the estimated $ASUT$ values for the aggregate spillover effect and for each of the two spillover dimensions, with standard errors clustered at the state and the unit levels. The estimated spillover effects are generally imprecise and are not statistically significant. The point estimates nevertheless display different patterns across the two dimensions. The non-geographical spillover estimates are positive in 2001 and negative from 2002 onward, while the geographical spillover estimates are positive throughout.

One possible interpretation of this temporal pattern is that the two spillover dimensions interact, as discussed above. Following the introduction of Lojack in a state, the state itself may become less attractive to thieves, leading some of them, over time, to shift their activity to nearby states rather than continue targeting unprotected models within the same state. Such a response could generate a gradual reduction in thefts of unprotected models within

	Linear			Poisson		
	Estimate	SE(st)	SE(id)	Estimate	SE(st)	SE(id)
$ASUT_{2001}$	0.923	0.569	0.469	0.338	0.944	1.011
$ASUT_{2002}$	-0.499	1.067	0.858	-0.251	1.349	1.120
$ASUT_{2003}$	-2.089	2.167	1.636	-1.322	1.735	1.362
$ASUT_{2004}$	-1.836	2.123	1.755	-1.773	1.887	1.577
$ASUT^{NG}_{2001}$	3.590	1.735	1.859	1.095	3.571	3.885
$ASUT^{NG}_{2002}$	-2.526	4.219	3.537	-1.455	5.215	4.395
$ASUT^{NG}_{2003}$	-10.579	8.320	7.455	-6.926	6.938	5.752
$ASUT^{NG}_{2004}$	-9.405	8.396	8.011	-9.104	7.517	6.656
$ASUT^{G}_{2001}$	0.159	0.130	0.170	0.139	0.193	0.205
$ASUT^{G}_{2002}$	0.014	0.220	0.160	0.018	0.216	0.165
$ASUT^{G}_{2003}$	0.266	0.348	0.190	0.233	0.328	0.184
$ASUT^{G}_{2004}$	0.264	0.351	0.206	0.261	0.369	0.214

Table 3: Estimates of the aggregate $ASUT$. The standard errors are clustered at the state (SE(st)) and the unit (SE(id)) levels. $ASUT_t$ denotes the aggregate spillover effect, while $ASUT_t^{NG}$ and $ASUT_t^{G}$ denote the non-geographical and geographical spillover effects, respectively.

Lojack-installed states while at the same time increasing theft activity in nearby untreated states through geographical displacement.

Lastly, we discuss the diagnostic tests described in Remarks 3 and 4. For the diagnostic test of Assumption 4 outlined in Remark 3, under the displacement mechanism emphasized by Gonzalez-Navarro [2013] we would expect that $ATT_0(g,t) < 0$ and $AST(g,t) \geq 0$. This is because the installation of Lojack would reduce auto thefts for the treated models, while potentially causing car thieves to redirect their efforts toward other car models, leading to a possible increase in theft attempts. We find that all estimated $ATT_0(g,t)$ values are negative (not reported here), not detecting evidence that $AST(g,t) > 0$.

For the diagnostic tests of Assumption 5 described in Remark 4, we do not find evidence of violation of Equation (5) for both the linear and Poisson models, while we find evidence of violation of Equation (6) for the linear model. The latter result suggests that the Poisson model is a more suitable specification for our empirical application. Further details of these

diagnostic tests are provided in the Appendix.

8 Conclusion

We establish identifying assumptions and estimation procedures for the ATT without interference in a DiD setting with staggered treatment adoption and spillovers. Aside from the canonical DiD assumptions, identification is established under the assumption that once a unit receives treatment, it is no longer influenced by the spillover effect. This means the unit forfeits any spillovers it may have previously received and remains unaffected by spillovers from subsequently treated groups. This assumption, which is likely to hold in many contexts, unifies the multiple definitions of the ATT, simplifying policy evaluation and aligning with the definition of ATT under SUTVA.

To estimate the ATT without interference, we extend the TWFE regression approach of Wooldridge [2025] to account for spillovers in linear and nonlinear settings. Our approach can be used to easily calculate the ATT's standard error. We then revisit Gonzalez-Navarro [2013], who studied the effects of installing an auto theft prevention device known as Lojack. Our approach leads to a slightly larger effect of the policy relative to the original contribution's specification.

Finally, our Monte Carlo analysis brings attention to the inherent bias-variance trade-off involved in addressing staggered treatment and especially spillovers. We compare three different estimators: the traditional TWFE estimator, which overlooks both staggered adoption and spillovers; the estimator of Wooldridge [2025], which considers staggered adoption but not spillovers; and our proposed estimator, which addresses both factors. Our estimator proves to be competitive in various scenarios.

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SUPPLEMENTARY MATERIAL

A Diagnostic Test for Assumption 5

In this section, we provide details on two diagnostic tests that can be used to check validity of Assumption 5. It consists of two equations, given by Equations (5) and (6), which lead to the following two diagnostic tests.

First, Equation (5) can be checked by the following placebo test:

1. Divide the group $(G_i = \infty, H_i = 1)$ into two subgroups, where the researcher believes that Assumption 5 holds for the first subgroup but may be violated for the second subgroup, and label the latter as $(G_i = \infty, H_i = 2)$. For instance, in Example 2, we could divide states in $G_i = \infty$ into two groups depending on whether they are located in the boundaries of Mexico or in the interior, and label the latter as $(G_i = \infty, H_i = 2)$.

2. For this group $(G_i = \infty, H_i = 2)$, define $S_{it} = 1$ for all $t \geq q$ and 0 otherwise.

3. Run the regression specified in Equation (8).

4. If Equation (5) holds, then $\gamma_{\infty,2,t}$ should be zero for all $t \geq q$.

This placebo test splits the group $(G_i = \infty, H_i = 1)$ into two subgroups and considers the second subgroup as if it were affected by spillover effects. Note that this is not a formal test of Equation (5). For example, the placebo test will not be rejected if both groups $H_i = 1, 2$ are equally affected by spillovers.

Second, Equation (6) can be checked using the standard pre-trend test, given that there are multiple pre-treatment periods $(q \geq 3)$:

1. Select an integer k such that $1 \leq k \leq q-2$, and redefine S_{it} as:

$$S_{it} = \begin{cases} 1 & \text{if } t \geq q-k \text{ and } D_{it} = 0 \text{ and } H_i = 0 \\ 0 & \text{otherwise} \end{cases}.$$

2. Run the regression specified in Equation (8).

3. If Equation (6) holds, then $\gamma_{g,0,t}$ should be zero for $q-k \leq t < q$.

This test is similar to the standard pre-trend test, but the pre-treatment periods in the presence of spillovers are characterized by $t < q$, i.e., periods in which all groups are untreated. By contrast, under SUTVA, the pre-treatment periods for each group are defined as periods in which each group is untreated.

In the remainder of this section, we discuss the results of these tests for our application in Section 7. We note that the dataset comes with two pre-treatment years, namely 1999 and 2000. Because of the small number of pre-treatment periods as well as the small number of units entering treatment in years 2001 and 2002, we perform a pre-trend test which focuses on the year 2000 considering units that enter treatment in 2003. We find that we do not reject the null hypothesis of parallel trends for this group for the Poisson model, with a p-value of 0.494, but we reject it for the linear model. For the placebo tests, among the units in the $(G_i = \infty, H_i = 1)$ group, we define the placebo group as units in the three states closer to Mexico City (Coahuila, Chiapas, and Tamaulipas), while the base control group consists of units in the remaining six states that are farther from Mexico City than the placebo states (Campeche, Yucatan, Quintana Roo, Chihuahua, Sonora, and Baja California). We find that we do not reject the null hypothesis of the placebo test for both models, with p-values of 0.414 for the linear model and 0.351 for the Poisson model.

B Monte Carlo Simulation

In this section, we study the finite sample properties of our estimator in a simulated dataset, highlighting the bias-variance trade-off of our approach. In the absence of spillover effects, our estimator is less efficient than conventional estimators that rule out spillover effects. However, in the presence of spillovers, the conventional estimators become biased. Given this bias-variance trade-off, when the sample size is small, the improvement in bias may not sufficiently offset the loss in precision.

We consider a balanced panel dataset over T periods, with either a simultaneous or staggered adoption design, starting with a pre-treatment period of $t = 1$. We consider M units in each extended group $g \in \tilde{\mathcal{G}} \equiv \{2, \ldots, T, (\infty, 0), (\infty, 1)\}$, meaning that we have a total of $N = (T + 1)M$ units in the dataset. We consider the following data generating process (DGP) that embeds Assumptions 1, 2, 4 and 5. Depending on the specification of the outcome model—linear or Poisson—we adapt the relevant assumption, replacing Assumption 5 with 5' as necessary. The DGP for the linear model is given by:

$$Y_{it} = \alpha_{G_i} + \delta_t + \beta_{it} D_{it} + (1 - D_{it}) \cdot \sum_{j=1}^{M} \gamma_{it}^{j} \cdot D_{jt} + \varepsilon_{it},$$

where γ_{it}^{j} represents the spillover effect from unit j to unit i. The DGP for the Poisson model is specified similarly with the link function $F(x) = \exp(x)$. We parametrize the DGP as follows.

- The M units in each group $g \in \{2, \ldots, T, (\infty, 0), (\infty, 1)\}$ are homogeneous, implying that $\beta_{it} = \beta_{G_i t}$ and $\gamma_{it}^{j} = \gamma_{G_i t}^{G_j}$.

- Fixed effects are set to $\alpha_g = 26 - g + 1$ for all groups except for $(\infty, 0)$ and $(\infty, 1)$, where $\alpha_{(\infty,0)} = \alpha_{(\infty,1)} = 26 - T + 1$. This reflects selection into treatment because the units with earlier treatment have larger unit fixed effects. In the case of the Poisson

model, we instead set $\alpha_g = \log(26 - g + 1)$.

- Common time effects are set to $\delta_t = \bar{\alpha} \times 0.1 \times ((t-1) + \sin(t))$, where $\bar{\alpha}$ is the average of the unit fixed effects across all groups. This specification involves a linear upward trend $(t-1)$ and a period-specific fluctuation modeled through $\sin(\cdot)$.

- The treatment effect is set to $\beta_{gt} = 0.5\alpha_g/t$. This effect is heterogeneous across groups and time periods, but homogeneous within a group. The effect gradually diminishes over time, with β_{gt} decreasing in t for each group g. The immediate effect β_{gg} is largest for group $g = 2$ with the highest α_g. This parametrizes sorting on gain since α_g also correlates with treatment timing.

- Spillover effects are set to $\gamma^h_{gt} = -\rho \cdot \beta_{gt}/U_t$, representing displacement effects, where U_t is the number of untreated units at time t except for those in group $(\infty, 1)$. That is, for each treated unit i, we consider a total spillover effect of $-\rho \cdot \beta_{G_i t}$, where $\rho \in [0, 1]$ denotes the spillover intensity. This total effect is then evenly spread among all untreated units excluding those in $(\infty, 1)$. As a result, each untreated unit receives a spillover effect of $-\rho \cdot \beta_{G_i t}/U_t$ from the treated unit i.

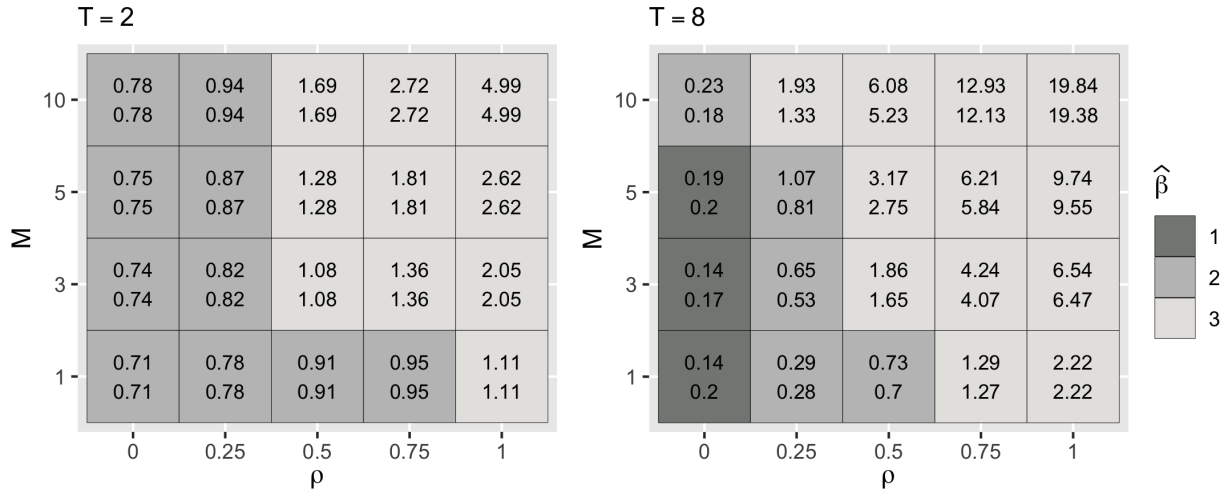
With this parametrization, Y_{it} is generated with an independent additive error term $\epsilon_{it} \sim N(0, \max(\alpha_{G_i})/10)$ for the linear model, and according to Poisson distribution for the Poisson model. We then estimate the aggregate ATT_0 defined as in Section 7, namely $ATT_0 = (1/\overline{G}) \sum_{g=2}^{T} \sum_{t=g}^{T} ATT_0(g,t)$, where $\overline{G} \equiv T(T-1)/2$ is the total number of treated group-time pairs in the dataset. We compare the mean Absolute Bias and the Mean Squared Error (MSE) across the following estimators:

($\widehat{\beta}_1$) The estimator from standard TWFE regression, which neither accounts for staggered treatment adoption nor for spillovers.

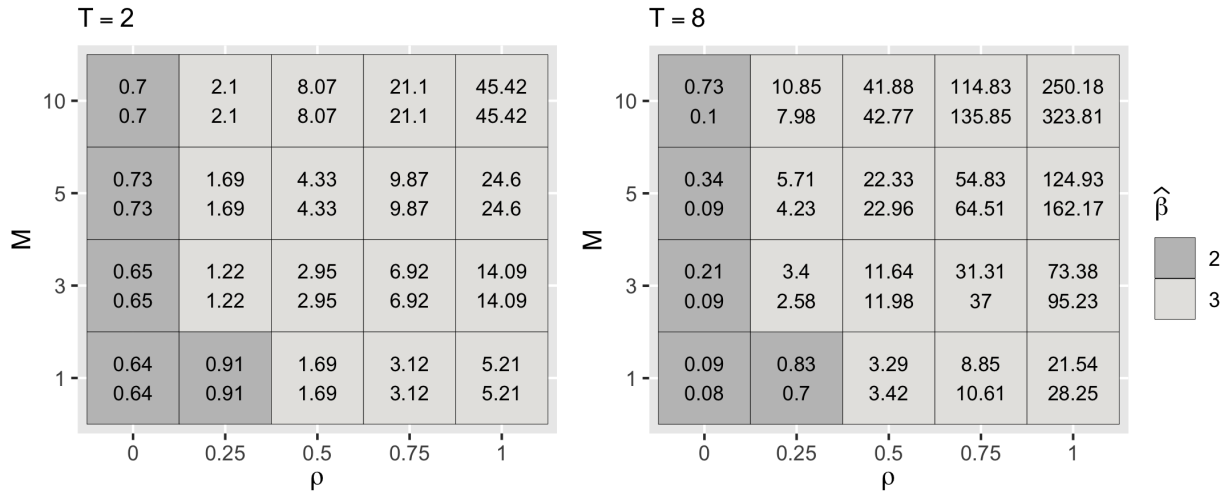
($\widehat{\beta}_2$) The estimator from fully interacted TWFE regression by Wooldridge [2025], which accounts for staggered treatment adoption but does not account for spillovers. This estimator is numerically equivalent to the imputation estimator by Borusyak et al. [2024].

($\widehat{\beta}_3$) Our estimator, which accounts for both staggered treatment adoption and spillovers.

Figure 3a and Table 4 present results from the linear DGP. The Figure visually contrasts the MSE across the three estimators to illustrate their relative performance under different scenarios, while the Table details their MSE and Absolute Bias values. Note that, when $T = 2$, the TWFE and the Wooldridge [2025] estimators are equivalent since treatment is not staggered. Overall, the relative performances of the estimators depend on the degree of spillovers, staggered treatment, and the number of units in each group. Intuitively, due to its efficiency, the TWFE has the lowest MSE in scenarios with no or little spillovers and with very few observations. As the number of observations increases and spillovers remain small, the Wooldridge [2025] estimator becomes the best-performing one, adjusting for staggered treatment without substantial bias. However, in scenarios where spillovers are not negligible and the number of units is large, our estimator achieves the lowest MSE, often by a large margin. Our estimator also performs better as treatment becomes more staggered ($T = 8$), highlighting our estimator's ability to accurately account for cumulative spillovers affecting the untreated units' outcomes. Furthermore, Figure 3b and Table 5 present results from the Poisson DGP, where our estimator performs even better relative to the TWFE and the Wooldridge [2025] ones.



(a) Linear



(b) Poisson

Figure 3: Comparison of the MSEs. Cell background color indicates the best-performing estimator. The numbers in cells represent the MSE ratios $\frac{MSE_1}{MSE_3}$ and $\frac{MSE_2}{MSE_3}$ respectively. The subscripts refer to: (1) TWFE estimator, (2) Wooldridge [2025] estimator, and (3) our estimator.

Table 4: MSE and Absolute Bias values - Linear

ρ	T	M	ATT	$\lvert Bias_1 \rvert$	$\lvert Bias_2 \rvert$	$\lvert Bias_3 \rvert$	MSE_1	MSE_2	MSE_3
0.00	2	1	-6.500	**3.589**	**3.589**	4.235	**19.642**	**19.642**	27.519
		3	-6.500	**2.060**	**2.060**	2.374	**6.613**	**6.613**	8.957
		5	-6.500	**1.615**	**1.615**	1.873	**3.991**	**3.991**	5.335
		10	-6.500	**1.159**	**1.159**	1.313	**2.134**	**2.134**	2.745
	8	1	-2.297	**0.898**	1.038	2.336	**1.244**	1.693	8.651
		3	-2.297	**0.554**	0.606	1.463	**0.477**	0.569	3.336
		5	-2.297	**0.463**	0.473	1.051	**0.334**	0.350	1.789
		10	-2.297	0.384	**0.334**	0.782	0.217	**0.172**	0.954
0.25	2	1	-6.500	**3.807**	**3.807**	4.321	**22.129**	**22.129**	28.292
		3	-6.500	**2.153**	**2.153**	2.391	**7.274**	**7.274**	8.856
		5	-6.500	**1.715**	**1.715**	1.824	**4.584**	**4.584**	5.280
		10	-6.500	**1.316**	**1.316**	1.342	**2.642**	**2.642**	2.805
	8	1	-2.297	1.403	**1.335**	2.456	2.776	**2.717**	9.587
		3	-2.297	1.396	**1.186**	1.509	2.314	**1.876**	3.561
		5	-2.297	1.355	1.116	**1.114**	2.054	**1.559**	1.917
		10	-2.297	1.342	1.069	**0.783**	1.928	1.324	**0.999**
0.50	2	1	-6.500	**3.870**	**3.870**	4.012	**23.616**	**23.616**	25.838
		3	-6.500	2.565	2.565	**2.395**	10.048	10.048	**9.274**
		5	-6.500	2.139	2.139	**1.877**	6.991	6.991	**5.448**
		10	-6.500	1.796	1.796	**1.338**	4.700	4.700	**2.787**
	8	1	-2.297	2.403	**2.284**	2.465	6.923	**6.688**	9.499
		3	-2.297	2.367	2.184	**1.440**	5.985	5.317	**3.220**
		5	-2.297	2.374	2.178	**1.087**	5.858	5.073	**1.847**
		10	-2.297	2.404	2.212	**0.782**	5.900	5.070	**0.970**
0.75	2	1	-6.500	**4.138**	**4.138**	4.162	**26.753**	**26.753**	28.210
		3	-6.500	2.905	2.905	**2.386**	12.336	12.336	**9.081**
		5	-6.500	2.652	2.652	**1.869**	9.928	9.928	**5.476**
		10	-6.500	2.405	2.405	**1.330**	7.524	7.524	**2.768**
	8	1	-2.297	3.448	3.346	**2.581**	13.089	12.935	**10.172**
		3	-2.297	3.452	3.352	**1.371**	12.300	11.811	**2.899**
		5	-2.297	3.417	3.290	**1.099**	11.884	11.162	**1.912**
		10	-2.297	3.424	3.308	**0.766**	11.839	11.106	**0.915**
1.00	2	1	-6.500	4.373	4.373	**4.148**	29.442	29.442	**26.621**
		3	-6.500	3.568	3.568	**2.355**	17.556	17.556	**8.570**
		5	-6.500	3.370	3.370	**1.906**	15.018	15.018	**5.723**
		10	-6.500	3.305	3.305	**1.270**	12.749	12.749	**2.555**
	8	1	-2.297	4.473	4.419	**2.453**	21.073	21.098	**9.491**
		3	-2.297	4.445	4.399	**1.394**	20.124	19.900	**3.076**
		5	-2.297	4.446	4.389	**1.134**	19.983	19.587	**2.051**
		10	-2.297	4.486	4.426	**0.802**	20.246	19.771	**1.020**

Note. Results over 1000 repetitions. Subscript refers to: (1) TWFE estimator, (2) Wooldridge [2025] estimator, and (3) our estimator. The lowest value across estimators is in bold.

Table 5: MSE and Absolute Bias values - Poisson

ρ	T	M	ATT	$\lvert Bias_1 \rvert$	$\lvert Bias_2 \rvert$	$\lvert Bias_3 \rvert$	MSE_1	MSE_2	MSE_3
0.00	2	1	-26.780	**9.701**	**9.701**	11.905	**153.431**	**153.431**	239.845
		3	-26.780	**5.540**	**5.540**	6.740	**47.762**	**47.762**	73.687
		5	-26.780	**4.245**	**4.245**	5.032	**29.219**	**29.219**	40.018
		10	-26.780	**3.071**	**3.071**	3.689	**14.968**	**14.968**	21.449
	8	1	-30.865	9.057	**8.085**	27.674	121.582	**105.769**	1347.861
		3	-30.865	8.280	**4.863**	15.381	85.855	**36.796**	404.870
		5	-30.865	8.044	**3.526**	11.869	75.668	**19.522**	224.123
		10	-30.865	8.125	**2.536**	7.890	71.494	**9.967**	98.496
0.25	2	1	-26.780	**12.067**	**12.067**	12.275	**248.172**	**248.172**	272.296
		3	-26.780	7.698	7.698	**6.889**	93.040	93.040	**75.998**
		5	-26.780	6.601	6.601	**5.072**	66.846	66.846	**39.553**
		10	-26.780	5.781	5.781	**3.857**	47.955	47.955	**22.876**
	8	1	-30.865	34.556	30.313	**28.662**	1266.576	**1058.689**	1518.343
		3	-30.865	33.742	28.917	**14.216**	1161.546	882.219	**341.538**
		5	-30.865	33.993	28.983	**11.307**	1169.549	866.469	**204.729**
		10	-30.865	33.858	28.888	**8.140**	1153.220	847.927	**106.282**
0.50	2	1	-26.780	15.622	15.622	**11.861**	390.349	390.349	**231.468**
		3	-26.780	12.878	12.878	**6.984**	231.040	231.040	**78.199**
		5	-26.780	11.946	11.946	**5.106**	184.497	184.497	**42.616**
		10	-26.780	12.066	12.066	**3.646**	167.718	167.718	**20.788**
	8	1	-30.865	67.194	67.826	**27.440**	4603.887	4782.310	**1398.162**
		3	-30.865	66.939	67.699	**15.029**	4508.284	4638.895	**387.225**
		5	-30.865	66.675	67.477	**11.203**	4462.200	4587.222	**199.809**
		10	-30.865	66.688	67.328	**8.195**	4456.398	4551.466	**106.420**
0.75	2	1	-26.780	22.597	22.597	**12.162**	762.129	762.129	**244.327**
		3	-26.780	20.320	20.320	**6.716**	503.553	503.553	**72.748**
		5	-26.780	19.418	19.418	**5.204**	430.610	430.610	**43.640**
		10	-26.780	20.115	20.115	**3.599**	430.218	430.218	**20.391**
	8	1	-30.865	108.579	118.317	**26.869**	11902.528	14259.553	**1344.294**
		3	-30.865	108.185	117.445	**15.289**	11742.232	13877.573	**375.027**
		5	-30.865	108.328	117.403	**11.473**	11757.889	13835.222	**214.460**
		10	-30.865	108.571	118.043	**7.979**	11799.580	13960.016	**102.759**
1.00	2	1	-26.780	31.420	31.420	**11.927**	1297.693	1297.693	**249.215**
		3	-26.780	30.005	30.005	**6.688**	1008.487	1008.487	**71.590**
		5	-26.780	30.104	30.104	**5.017**	972.364	972.364	**39.530**
		10	-26.780	29.547	29.547	**3.566**	906.943	906.943	**19.970**
	8	1	-30.865	164.055	187.416	**27.132**	27086.351	35529.966	**1257.582**
		3	-30.865	163.392	186.002	**15.000**	26748.599	34712.731	**364.521**
		5	-30.865	163.930	186.676	**11.684**	26902.365	34921.680	**215.341**
		10	-30.865	163.562	186.035	**8.031**	26768.157	34646.567	**106.997**

Note. Results over 1000 repetitions. Subscript refers to: (1) TWFE estimator, (2) Wooldridge [2025] estimator, and (3) our estimator. The lowest value across estimators is in bold.

C Proofs

C.1 Proof of Proposition 1

Under Assumptions 2 and 3, for each group g at time t, we can express Y_{it} as

$$\begin{aligned} Y_{it} &= Y_{it}(G_i, \mathbf{G}_{-i}) \\ &= Y_{it}(\infty, \infty_{-i}) + [Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})] + [Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i})] \\ &= \alpha_i + \delta_t + [Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})] + [Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i})] + \varepsilon_{it}, \end{aligned}$$

where the last equality follows from Remark 1, in which $\mathbb{E}(\varepsilon_{it}|G_i = g) = 0$ for every group g at time t. Define

$$\beta_{it} = Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i}),$$

$$\gamma_{it} = Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i}).$$

We can then simplify the expression for Y_{it} as

$$Y_{it} = \alpha_i + \delta_t + \beta_{it} + \gamma_{it} + \varepsilon_{it}.$$

In this expression, the parameter of interest $ATT_0(g,t)$ for $t \geq g$ is given by

$$ATT_0(g,t) = \mathbb{E}(\beta_{it}|G_i = g),$$

and $AST(g,t)$ for $t \geq g$ is given by

$$AST(g,t) = \mathbb{E}(\gamma_{it}|G_i = g).$$

Using these expressions, for every group $g \in \mathcal{G}$ such that $2 \leq g < \infty$ and time $t \geq g$, we can write the expectation of Y_{it} as

$$\mathbb{E}(Y_{it}|G_i = g) = \mathbb{E}(\alpha_i|G_i = g) + \delta_t + ATT_0(g,t) + AST(g,t), \tag{17}$$

where we used $\mathbb{E}(\varepsilon_{it}|G_i = g) = 0$.

Now we show that $ATT_0(g,t)$ is identified if and only if $\delta_t + AST(g,t)$ is identified, for every group $g \in \mathcal{G}$ such that $2 \leq g < \infty$ and time $t \geq g$. First, suppose that $\delta_t + AST(g,t)$ is identified. Let d_0 be the identified value. Then we can rewrite Equation (17) as

$$\mathbb{E}(Y_{it}|G_i = g) = \mathbb{E}(\alpha_i|G_i = g) + d_0 + ATT_0(g,t). \tag{18}$$

Now we show that $\mathbb{E}(\alpha_i|G_i = g)$ is identified from the data at $t = 1$. Note first that, under the assumptions of Proposition 1, all units are untreated at $t = 1$. This implies that

$$Y_{i1} = Y_{i1}(G_i, \mathbf{G}_{-i}) = Y_{i1}(\infty, \infty_{-i}) = \alpha_i + \delta_1 + \varepsilon_{i1}$$

by Assumptions 2 and 3. Then it follows that

$$\mathbb{E}(Y_{i1}|G_i = g) = \mathbb{E}(\alpha_i + \delta_1 + \varepsilon_{i1}|G_i = g) = \mathbb{E}(\alpha_i|G_i = g), \tag{19}$$

where $\delta_1 = 0$ and $\mathbb{E}(\varepsilon_{it}|G_i = g) = 0$ by Remark 1. We can then rewrite Equation (18) as

$$ATT_0(g,t) = \mathbb{E}(Y_{it}|G_i = g) - d_0 - \mathbb{E}(Y_{i1}|G_i = g),$$

which shows that $ATT_0(g,t)$ is identified because $\mathbb{E}(Y_{it})$ and $\mathbb{E}(Y_{i1})$ are identifiable whenever $g \in \mathcal{G}$, i.e., whenever the group appears in the data.

Conversely, suppose that $ATT_0(g,t)$ is identified. Let b_0 be the identified value. Then we

can rewrite Equation (17) as

$$\mathbb{E}(Y_{it}|G_i = g) = \mathbb{E}(\alpha_i|G_i = g) + \delta_t + b_0 + AST(g,t).$$

Using Equation (19), we can write

$$\delta_t + AST(g,t) = \mathbb{E}(Y_{it}|G_i = g) - b_0 - \mathbb{E}(Y_{i1}|G_i = g),$$

which shows that $\delta_t + AST(g,t)$ is identified. ■

C.2 Proof of Theorem 1

We first prove identification of $ATT_0(g,t)$. Recall that, for $t \geq g$:

$$ATT_0(g,t) = \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = g),$$

which we can rewrite as

$$\begin{aligned} ATT_0(g,t) = \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g) \\ - \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g) \\ - \mathbb{E}(Y_{it}(\infty_i, \infty_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g). \end{aligned}$$

We now simplify each term in the right-hand side. The first term simplifies to:

$$\mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = g).$$

In addition, the second term is zero for $t \geq g$ by Assumption 4:

$$\mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g) = 0.$$

Lastly, using Assumption 5, the third term simplifies to:

$$\begin{aligned}
&\mathbb{E}(Y_{it}(\infty_i, \infty_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g) \\
&= \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1) \\
&= \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).
\end{aligned}$$

Therefore, it follows that

$$ATT_0(g,t) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = g) - \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1)$$

for $t \geq g$.

Next, we prove identification of $ASUT(g,t)$. Recall that, for $q \leq t < g$:

$$ASUT(g,t) = \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = g),$$

which we can rewrite as

$$\begin{aligned}
ASUT(g,t) = \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g) \\
- \mathbb{E}(Y_{it}(\infty_i, \infty_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g).
\end{aligned}$$

We now simplify each term in the right-hand side. Using Assumption 2, the first term simplifies to:

$$\begin{aligned}
\mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g) &= \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{i1}(g, \mathbf{G}_{-i})|G_i = g) \\
&= \mathbb{E}(Y_{it} - Y_{i1}|G_i = g).
\end{aligned}$$

In addition, using Assumptions 2 and 5, the second term simplifies to:

$$\begin{aligned}
&\mathbb{E}(Y_{it}(\infty_i, \infty_{-i}) - Y_{i1}(\infty_i, \infty_{-i})|G_i = g) \\
&= \mathbb{E}(Y_{it}(\infty, \mathbf{G}_{-i}) - Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1) \\
&= \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).
\end{aligned}$$

Therefore, it follows that

$$ASUT(g,t) = \mathbb{E}(Y_{it} - Y_{i1}|G_i = g) - \mathbb{E}(Y_{it} - Y_{i1}|G_i = \infty, H_i = 1).$$

for $q \leq t < g$. ■

C.3 Proof of Theorem 2

Using Remark 5, we can express Y_{it} for every $2 \leq g < \infty$ as:

$$\begin{aligned}
&\mathbb{E}(Y_{it}|G_i = g) \\
&= \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i})|G_i = g) \\
&= \mathbb{E}(Y_{it}(\infty, \infty_{-i}) + [Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})] + [Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})]|G_i = g) \\
&= \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = g) + \delta_t + \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g) \\
&\qquad + \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g).
\end{aligned}$$

Since $G_i \neq \infty$ implies $H_i = 0$, we can rewrite the above as

$$\begin{aligned}
\mathbb{E}(Y_{it}|G_i = g, H_i = 0) = \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = g, H_i = 0) + \delta_t \\
+ \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g, H_i = 0) \\
+ \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g, H_i = 0).
\end{aligned} \tag{20}$$

In addition, using Remark 5, we can express Y_{it} for $G_i = \infty$ and $H_i = 1$ as:

$$\begin{aligned}
&\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1) \\
&= \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i})|G_i = \infty, H_i = 1) \\
&= \mathbb{E}(Y_{it}(\infty, \infty_{-i}) + [Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})]|G_i = \infty, H_i = 1) \\
&= \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1) + \delta_t.
\end{aligned} \tag{21}$$

Lastly, Remark 5 and the definition of δ_t respectively implies that:

$$\begin{aligned}
&\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = \infty) = \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = \infty) + \delta_t, \\
&\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = \infty, H_i = 1) = \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 1) + \delta_t,
\end{aligned}$$

which implies that

$$\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = \infty, H_i = 0) = \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 0) + \delta_t.$$

Therefore, we can express Y_{it} for $G_i = \infty$ and $H_i = 0$ as:

$$\begin{aligned}
&\mathbb{E}(Y_{it}|G_i = \infty, H_i = 0) \\
&= \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i})|G_i = \infty, H_i = 0) \\
&= \mathbb{E}(Y_{it}(\infty, \infty_{-i}) + [Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})]|G_i = \infty, H_i = 0) \\
&= \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = \infty, H_i = 0) + \delta_t \\
&\qquad + \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = \infty, H_i = 0).
\end{aligned} \tag{22}$$

Now, let

$$\begin{aligned}
A_{gh} &\equiv \mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = g, H_i = h), \\
B_{ght} &\equiv \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g, H_i = h), \\
C_{ght} &\equiv \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g, H_i = h).
\end{aligned}$$

We can rewrite Equations (20), (21) and (22) as:

$$\begin{aligned}
\mathbb{E}(Y_{it}|G_i = g, H_i = 0) &= A_{gh} + \delta_t + B_{ght} + C_{ght} \quad \text{if } 2 \leq g < \infty, \\
\mathbb{E}(Y_{it}|G_i = \infty, H_i = 0) &= A_{\infty,h} + \delta_t + C_{\infty,h,t}, \\
\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1) &= A_{\infty,h} + \delta_t.
\end{aligned}$$

In addition, by the definitions of B_{ght} and C_{ght} and Assumptions 2 and 4, for $2 \leq g < \infty$, it follows that

$$B_{ght} = 0 \text{ if } t < g,$$

$$C_{ght} = 0 \text{ if } t < q \quad \text{or} \quad g \leq t,$$

and $C_{\infty,h,t} = 0$ if $t < q$. Therefore, we can further rewrite Equations (20), (21) and (22) as:

$$\mathbb{E}(Y_{it}|G_i = g, H_i = h) = \begin{cases} A_{gh} + \delta_t & \text{if } t < q \\ A_{gh} + \delta_t & \text{if } g = \infty, h = 1 \text{ and } q \leq t \\ A_{gh} + \delta_t + C_{\infty,h,t} & \text{if } g = \infty, h = 0 \text{ and } q \leq t \\ A_{gh} + \delta_t + C_{ght} & \text{if } 2 \leq g < \infty \text{ and } q \leq t < g \\ A_{gh} + \delta_t + B_{ght} & \text{if } 2 \leq g < \infty \text{ and } t \geq g \end{cases}.$$

Note that, according to the definition of D_{it} in the main text, D_{it} is equal to 1 if $2 \leq G_i < \infty$ and $t \geq G_i$. Similarly, according to the definition in the main text, S_{it} is equal to 1 if $(G_i = \infty, H_i = 0)$ and $q \leq t$ holds or if $2 \leq G_i < \infty$ and $q \leq t < g$. Then we can combine the above expressions into one unified expression as follows:

$$\begin{aligned}
\mathbb{E}(Y_{it}|G_i = g, H_i = h) = A_{gh} + \delta_t &+ \sum_{g',h'} \sum_{t'=g}^{T} B_{g'h't'} \mathbf{1}(g' = g, h' = h, t' = t) D_{it} \\
&+ \sum_{g',h'} \sum_{t'=q}^{\min\{g'-1,T\}} C_{g'h't'} \mathbf{1}(g' = g, h' = h, t' = t) S_{it}.
\end{aligned} \tag{23}$$

Now we proceed to prove the theorem. Let $\mathbf{X}_{it}$ be the vector of regressors in Equa-

tion (9), namely indicators of (g,h), indicators of t, interactions between D_{it} and indicators of (g,h,t), and interactions between S_{it} and indicators of (g,h,t). Then the regressors $\mathbf{X}_i \equiv (\mathbf{X}_{i1},\ldots,\mathbf{X}_{iT})$ identify the extended group label (g,h) and vice versa, because the interactions of D_{it} identifies the original group label $g \in \mathcal{G}$ and the interactions of S_{it} distinguishes $(\infty,0)$ and $(\infty,1)$. This implies that

$$\mathbb{E}(Y_{it}|\mathbf{X}_i) = \mathbb{E}(Y_{it}|G_i = g(\mathbf{X}_i), H_i = h(\mathbf{X}_i))$$

where $g(\mathbf{X}_i)$ and $h(\mathbf{X}_i)$ are the extended group label identified by $\mathbf{X}_i$. Then we can write Equation (23) as:

$$\begin{aligned}\mathbb{E}(Y_{it}|\mathbf{X}_i) = A_{g(\mathbf{X}_i)h(\mathbf{X}_i)} + \delta_t \\ + \sum_{g',h'} \sum_{t'=g(\mathbf{X}_i)}^{T} B_{g'h't'} \mathbf{1}(g' = g(\mathbf{X}_i), h' = h(\mathbf{X}_i), t' = t) D_{it} \\ + \sum_{g',h'} \sum_{t'=q}^{\min\{g'-1,T\}} C_{g'h't'} \mathbf{1}(g' = g(\mathbf{X}_i), h' = h(\mathbf{X}_i), t' = t) S_{it},\end{aligned}$$

which coincides with the population regression of Equation (9). Then it follows that

$$\begin{aligned}\beta_{gt} &= B_{g,0,t} \quad \text{for } 2 \le g < \infty \text{ and } t \ge g, \\ \gamma_{g,0,t} &= C_{g,0,t} \quad \text{for } q \le t < g.\end{aligned}$$

which means that

$$\begin{aligned}\beta_{gt} &= \mathbb{E}(Y_{it}(g,\infty_{-i}) - Y_{it}(\infty,\infty_{-i})|G_i = g) \quad \text{for } 2 \le g < \infty \text{ and } t \ge g, \\ \gamma_{g,0,t} &= \mathbb{E}(Y_{it}(\infty_i,\mathbf{G}_{-i}) - Y_{it}(\infty_i,\infty_{-i})|G_i = g, H_i = 0) \quad \text{for } q \le t < g,\end{aligned}$$

by the fact that $G_i \neq \infty$ implies $H_i = 0$ and by Assumption 2. ■

C.4 Proof of Corollary 1

Similarly to the proof of Proposition 1, for each group g at time t, we can express Y_{it} as

$$\begin{aligned} Y_{it} &= Y_{it}(G_i, \mathbf{G}_{-i}) \\ &= Y_{it}(\infty, \infty_{-i}) + [Y_{it}(G_i, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})] + [Y_{it}(G_i, \mathbf{G}_{-i}) - Y_{it}(G_i, \infty_{-i})]\,. \end{aligned}$$

With an abuse of notation, let $\alpha_g \equiv F(\mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = g))$ and $\delta_t \equiv F(\mathbb{E}(Y_{it}(\infty, \infty_{-i})|G_i = \infty)) - F(\mathbb{E}(Y_{i1}(\infty, \infty_{-i})|G_i = \infty))$. We can then write the expectation of Y_{it} as

$$\mathbb{E}(Y_{it}|G_i = g) = F^{-1}(\alpha_g + \delta_t) + ATT_0(g,t) + AST(g,t),$$

where

$$ATT_0(g,t) = \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty, \infty_{-i})|G_i = g),$$

and

$$AST(g,t) = \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g).$$

Then, by replicating the arguments in Proposition 1 that starts from Equation (17), it is straightforward to show that $ATT_0(g,t)$ is identified if and only if $F^{-1}(\alpha_g + \delta_t) + AST(g,t)$ is identified. ■

C.5 Proof of Corollary 2

Similarly to Theorem 1, we first prove identification of $ATT_0(g,t)$. Recall that, for $t \geq g$:

$$ATT_0(g,t) = \mathbb{E}(Y_{it}(g, \infty_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = g),$$

which we can rewrite as

$$ATT_0(g,t) = \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = g) \\ - \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g).$$

We now simplify each term in the right-hand side. Using Remark 9 and Assumption 2, the first term simplifies to:

$$\begin{aligned} &\mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = g) \\ &= \mathbb{E}(Y_{it}|G_i = g) - \mathbb{E}(Y_{it}(\infty_i, \infty_{-i})|G_i = g) \\ &= \mathbb{E}(Y_{it}|G_i = g) - F^{-1}\Big[F(\mathbb{E}(Y_{i1}|G_i = g)) + \\ &\qquad F(\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)) - F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1))\Big]. \end{aligned}$$

In addition, the second term is zero for $t \geq g$ by Assumption 4:

$$\mathbb{E}(Y_{it}(g, \mathbf{G}_{-i}) - Y_{it}(g, \infty_{-i})|G_i = g) = 0.$$

Therefore, it follows that

$$\begin{aligned} ATT_0(g,t) = \mathbb{E}(Y_{it}|G_i = g) - F^{-1}\Big[&F(\mathbb{E}(Y_{i1}|G_i = g)) + \\ &F(\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)) - F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1))\Big] \end{aligned}$$

for $t \geq g$.

Next, we prove identification of $ASUT(g,t)$ for $g \in \mathcal{G}\backslash\{\infty\}$. Recall that, for $q \leq t < g$:

$$ASUT(g,t) = \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = g).$$

which we can rewrite as

$$ASUT(g,t) = \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i})|G_i = g) - \mathbb{E}(Y_{it}(\infty_i, \infty_{-i})|G_i = g).$$

We now simplify each term in the right-hand side. Using Assumption 2, the first term simplifies to:

$$\begin{aligned}\mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i})|G_i = g) &= \mathbb{E}(Y_{it}(g, \mathbf{G}_{-i})|G_i = g) \\ &= \mathbb{E}(Y_{it}|G_i = g).\end{aligned}$$

In addition, using Remark 9 and Assumption 2, the second term simplifies to:

$$\begin{aligned}\mathbb{E}(Y_{it}(\infty_i, \infty_{-i})|G_i = g) = F^{-1}\Big[&F(\mathbb{E}(Y_{i1}|G_i = g)) + \\ &F(\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)) - F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1))\Big].\end{aligned}$$

Therefore, it follows that

$$\begin{aligned}ASUT(g,t) = \mathbb{E}(Y_{it}|G_i = g) - F^{-1}\Big[&F(\mathbb{E}(Y_{i1}|G_i = g)) + \\ &F(\mathbb{E}(Y_{it}|G_i = \infty, H_i = 1)) - F(\mathbb{E}(Y_{i1}|G_i = \infty, H_i = 1))\Big].\end{aligned}$$

for $q \leq t < g$. Lastly, for $ASUT(\infty, 0, t)$, defined by

$$ASUT(\infty, 0, t) = \mathbb{E}(Y_{it}(\infty_i, \mathbf{G}_{-i}) - Y_{it}(\infty_i, \infty_{-i})|G_i = \infty, H_i = 0),$$

identification follows from the same argument used for $ASUT(g,t)$ for $g \in \mathcal{G}\backslash\{\infty\}$, by conditioning on $(G_i = \infty, H_i = 0)$ and using Equation (15). ■